

DATA-DRIVEN DETECTION OF RICE LEAF DISEASES USING YOLO
ALGORITHMS

AYAZ RAHMAN BHUIYAN

UNIVERSITI TEKNOLOGI MALAYSIA



UNIVERSITI TEKNOLOGI MALAYSIA
DECLARATION OF Choose an item.

Author's full name : AYAZ RAHMAN BHUIYAN

Student's Matric No. : MCS231023 Academic Session : 2023-2024

Date of Birth : 26/09/1998 UTM Email : Rahman.1998@graduate.utm.my

Choose an item. Title : DATA-DRIVEN DETECTION OF RICE LEAF DISEASES USING YOLO ALGORITHMS

I declare that this thesis is classified as:

☒

OPEN ACCESS

I agree that my report to be published as a hard copy or made available through online open access.

☐

RESTRICTED

Contains restricted information as specified by the organization/institution where research was done.
(The library will block access for up to three (3) years)

☐

CONFIDENTIAL

Contains confidential information as specified in the Official Secret Act 1972)

(If none of the options are selected, the first option will be chosen by default)

I acknowledged the intellectual property in the Choose an item. belongs to Universiti Teknologi Malaysia, and I agree to allow this to be placed in the library under the following terms :

1. This is the property of Universiti Teknologi Malaysia
2. The Library of Universiti Teknologi Malaysia has the right to make copies for the purpose of only.
3. The Library of Universiti Teknologi Malaysia is allowed to make copies of this Choose an item. for academic exchange.

Signature of Student:

Signature:

Full Name: Ayaz Rahman Bhuiyan

Date : 17/06/24

Approved by Supervisor(s)

Signature of Supervisor I:

Signature of Supervisor II

Full Name of Supervisor I
 PROF MADYA DR. SITI ZAITON BT.
 MOHD HASHIM

Full Name of Supervisor II

Date :

Date :

NOTES : If the thesis is CONFIDENTIAL or RESTRICTED, please attach with the letter from the organization with period and reasons for confidentiality or restriction

This letter should be written by a supervisor and addressed to Perpustakaan UTM. A copy of this letter should be attached to the thesis.

Date:

Librarian

Jabatan Perpustakaan UTM,
Universiti Teknologi Malaysia,
Johor Bahru, Johor

Sir,

CLASSIFICATION OF THESIS AS RESTRICTED/CONFIDENTIAL

TITLE: Click or tap here to enter text.

AUTHOR'S FULL NAME: Click or tap here to enter text.

Please be informed that the above-mentioned thesis titled _____ should be classified as RESTRICTED/CONFIDENTIAL for a period of three (3) years from the date of this letter. The reasons for this classification are

- (i)
- (ii)
- (iii)

Thank you.

Yours sincerely,

SIGNATURE:

NAME:

ADDRESS OF SUPERVISOR:

“I hereby declare that I have read this Choose an item. and in Choose an item.
opinion this Choose an item. is sufficient in term of scope and quality for the
award of the degree of Choose an item.”

Signature : _____
Name of Supervisor I : _____
Date : _____

Signature : _____
Name of Supervisor II : _____
Date : _____

Signature : _____
Name of Supervisor III : _____
Date : _____

Declaration of Cooperation

This is to confirm that this research has been conducted through a collaboration [Click or tap here to enter text.](#) and [Click or tap here to enter text.](#)

Certified by:

Signature :

Name :

Position :

Official Stamp

Date

* This section is to be filled up for theses with industrial collaboration

Pengesahan Peperiksaan

Tesis ini telah diperiksa dan diakui oleh:

Nama dan Alamat Pemeriksa Luar :

Nama dan Alamat Pemeriksa Dalam :

Nama Penyelia Lain (jika ada) :

Disahkan oleh Timbalan Pendaftar di Fakulti:

Tandatangan :

Nama :

Tarikh :

DATA-DRIVEN DETECTION OF RICE LEAF DISEASES USING YOLO ALGORITHMS

AYAZ RAHMAN BHUIYAN

A proposal submitted in partial fulfilment of the
requirements for the award of the degree of
Master of Data Science

Choose an item.
Faculty of Computing
Universiti Teknologi Malaysia

JUNE 2024

DECLARATION

I declare that this proposal entitled “*DATA-DRIVEN DETECTION OF RICE LEAF DISEASES USING YOLO ALGORITHMS*” is the result of my own research except as cited in the references. The proposal has not been accepted for any degree and is not concurrently submitted in candidature of any other degree.

Signature : 
.....
Name : AYAZ RAHMAN BHUIYAN
Date : 17 JUNE 2024

ACKNOWLEDGEMENT

I would like to express my deepest gratitude to my supervisor, Prof Madya Dr. Siti Zaiton Bt. Mohd Hashim, for her invaluable guidance, support, and encouragement throughout this research. Her insights and expertise have been instrumental in the successful completion of this thesis.

I extend my sincere thanks to my fellow researchers and colleagues, whose collaboration and constructive feedback have greatly contributed to this work. I am also grateful to my friends and family for their unwavering support and guidance during the challenging times of this journey. Their belief in me has been a constant source of motivation.

ABSTRACT

Rice leaf diseases significantly impact crop yields and food security, necessitating precise detection and management strategies. Traditional manual inspection methods are labor-intensive and prone to inaccuracies, particularly with complex natural images due to overlapping leaves and similar backgrounds. This research utilizes various versions of the You Only Look Once (YOLO) algorithm framework, specifically YOLOv8, YOLOv10, and YOLOv11, to compare the accuracy and efficiency of detecting rice leaf diseases. The study consists of four primary phases: problem finding and planning, data collection and labeling, training and custom model development, and model testing and analysis. Initially, a thorough literature review was conducted to identify existing gaps and challenges in the detection of rice leaf diseases. The research focused on three major rice diseases: bacterial blight, blast, and brown spot, utilizing comprehensive datasets sourced from various repositories. In the data collection phase, images were annotated using polygon labeling to enhance the disease-specific features accurately. Custom models were developed and evaluated based on their performance metrics, including detection accuracy, speed, and the confusion matrix. Among the models, YOLOv11 achieved the highest mean average precision (mAP) of 92.9%, followed by YOLOv10 at 89.9% and YOLOv8 at 88.0%. The results demonstrated that the enhanced YOLO-based models significantly outperformed traditional methods. The study highlights the potential of advanced machine learning techniques in revolutionizing agricultural practices, offering robust, real-time diagnostic tools for early disease management. This advancement contributes to sustainable crop management, ensuring higher productivity and better disease intervention strategies, thereby supporting global food security.

ABSTRAK

Penyakit daun padi memberi kesan ketara kepada hasil tanaman dan keselamatan makanan, memerlukan pengesanan dan strategi pengurusan yang tepat. Kaedah pemeriksaan manual tradisional adalah intensif buruh dan terdedah kepada ketidaktepatan, terutamanya dengan imej semula jadi yang kompleks disebabkan oleh daun yang bertindih dan latar belakang yang serupa. Penyelidikan ini menggunakan pelbagai versi rangka kerja algoritma Anda Hanya Lihat Sekali (YOLO), khususnya YOLOv8, YOLOv10, dan YOLOv11, untuk membandingkan ketepatan dan kecekapan pengesanan penyakit daun padi. Kajian ini terdiri daripada empat fasa utama: mencari dan merancang masalah, pengumpulan dan pelabelan data, latihan dan pembangunan model tersuai, dan ujian dan analisis model. Pada mulanya, kajian literatur yang menyeluruh telah dijalankan untuk mengenal pasti jurang dan cabaran sedia ada dalam pengesanan penyakit daun padi. Penyelidikan memfokuskan kepada tiga penyakit utama padi: penyakit hawar bakteria, letupan, dan bintik coklat, menggunakan set data komprehensif yang diperoleh daripada pelbagai repositori. Dalam fasa pengumpulan data, imej telah diberi anotasi menggunakan pelabelan poligon untuk meningkatkan ciri khusus penyakit dengan tepat. Model tersuai telah dibangunkan dan dinilai berdasarkan metrik prestasi mereka, termasuk ketepatan pengesanan, kelajuan dan matriks kekeliruan. Antara model, YOLOv11 mencapai purata ketepatan purata (mAP) tertinggi sebanyak 92.9%, diikuti oleh YOLOv10 pada 89.9% dan YOLOv8 pada 88.0%. Keputusan menunjukkan bahawa model berasaskan YOLO yang dipertingkatkan dengan ketara mengatasi kaedah tradisional. Kajian itu menyerlahkan potensi teknik pembelajaran mesin lanjutan dalam merevolusikan amalan pertanian, menawarkan alat diagnostik masa nyata yang mantap untuk pengurusan penyakit awal. Kemajuan ini menyumbang kepada pengurusan tanaman yang mampan, memastikan produktiviti yang lebih tinggi dan strategi intervensi penyakit yang lebih baik, sekali gus menyokong keselamatan makanan global.

TABLE OF CONTENTS

	TITLE	PAGE
	DECLARATION	iii
	ACKNOWLEDGEMENT	v
	ABSTRACT	vi
	ABSTRAK	vii
	TABLE OF CONTENTS	viii
	LIST OF TABLES	xii
	LIST OF FIGURES	xiii
	LIST OF ABBREVIATIONS	xv
	LIST OF SYMBOLS	xvi
	LIST OF APPENDICES	xvii
CHAPTER 1	INTRODUCTION	1
1.1	Introduction	1
1.2	Problem Background	2
1.3	Problem Statement	3
1.4	Research Objectives	3
1.5	Scopes of the project	4
1.6	Expected output	4
CHAPTER 2	LITERATURE REVIEW	7
2.1	Introduction	7
2.2	Overview of the rice leaf diseases	7
2.2.1	Bacterial Blight	8
2.2.2	Brown Spot	9
2.2.3	Rice Blast	10
2.3	Overall Impact and Challenges	11
2.4	Existing Rice Diseases Detection Method	12
2.4.1	Traditional methods of disease detection	12

	2.4.2	Advanced methods of disease detection	13
2.5		Concept of YOLO	14
	2.5.1	Evaluation of YOLO	15
	2.5.2	YOLO v8 16	
	2.5.3	YOLO v10	16
	2.5.4	YOLO v11	17
	2.5.5	Application of YOLO in Crop Disease Detection:	18
2.6		Enhanced Feature Selection	18
2.7		Related works	20
CHAPTER 3		RESEARCH METHODOLOGY	25
3.1		Introduction	25
3.2		Project Framework	25
	3.2.1	Phase 1: Problem Finding and Planning	26
	3.2.2	Phase 2: Data Collection and Analysis	27
	3.2.3	Phase 3: Train and Develop Custom Models	31
	3.2.4	Phase 4: Model Testing and Analysis	31
	3.2.5	Phase 5: Dashboard Development	32
	3.2.6	Algorithm Setup	33
	3.2.7	Software and Hardware Requirements	34
3.3		Chapter Summary	34
CHAPTER 4		EXPLORATORY DATA ANALYSIS	36
4.1		Introduction	36
4.2		Data Collection:	36
4.3		Exploratory Data Analysis:	37
CHAPTER 5		MODEL DEVELOPMENT	42
5.1		Introduction	42
5.2		Development of the Models:	42
	5.2.1	Training Data:	42
	5.2.2	Training Process:	43
	5.2.3	Evaluation Metrics:	45

5.3	YOLOv8 Model	45
5.4	YOLOv10 Model	48
5.5	YOLOv11 Model	50
5.6	Summary	53
CHAPTER 6	RESULTS AND DASHBOARD	55
6.1	Introduction	55
6.2	Results based on the YOLOv8 Model	55
6.2.1	Precision	55
6.2.2	Precision Recall	56
6.2.3	F1-Confidence Curve	57
6.2.4	Confusion Metrics	58
6.2.5	Loss Metrics	59
6.2.6	Results on Testing Images using the YOLOv8 Model	60
6.3	Results based on the YOLOv10 Model	63
6.3.1	Precision	63
6.3.2	Precision Recall	64
6.3.3	F1-Confidence Curve	65
6.3.4	Confusion Metrics	66
6.3.5	Loss Metrics	67
6.3.6	Results on Testing Images using the YOLOv8 Model	68
6.4	Results based on the YOLOv11 Model	71
6.4.1	Precision	71
6.4.2	Precision Recall	72
6.4.3	F1-Confidence Curve	73
6.4.4	Confusion Metrics	74
6.4.5	Loss Metrics	75
6.4.6	Results on Testing Images using the YOLOv11 Model	76
6.5	Comparisons of the Models:	79
6.6	Development and Analysis of Dashboard	80
6.6.1	Introduction to Dashboard	80

6.6.2	Dashboard Summary	80
6.7	Chapter Summary	82
CHAPTER 7	CONCLUSION AND RECOMMENDATIONS	85
7.1	Introduction	85
7.2	Summary of the Key Findings:	85
7.3	Contributions to Knowledge	86
7.4	Recommendations for Future Work	87
	REFERENCES	89

LIST OF TABLES

TABLE NO.	TITLE	PAGE
Table 2.1	Summary of related works	21
Table 3.1	Details of the datasets	28
Table 3.2	Software and hardware requirements	34
Table 3.1	Details of the datasets	36

LIST OF FIGURES

FIGURE NO.	TITLE	PAGE
Figure 2.1	(a), (b), (c) shows leaves with diseases. (d) shows healthy leaf	8
Figure 2.2	Bacterial blight disease on rice leaf	9
Figure 2.3	Brown spot disease on rice leaf	10
Figure 2.4	Blasts disease on rice leaf	11
Figure 2.5	Model architecture of YOLO (Redmon et al., 2015)	15
Figure 2.6	Evaluation of YOLO versions	16
Figure 2.7	Part (a) shows traditional object annotations, and part (b) shows polygon object annotations	19
Figure 3.1	Project Framework	26
Figure 3.2	Polygonal annotation and labelling of the images of Roboflow	30
Figure 4.1	Dataset classes overview	37
Figure 4.2	Dataset images overview	38
Figure 4.3	Dataset class balance	38
Figure 4.4	Dataset dimension insights	39
Figure 4.5	Annotation Heatmap	40
Figure 4.6	Histogram of object count by image	40
Figure 5.1	Flowchart of the models training process	44
Figure 5.2	Checking GPU availability	45
Figure 5.3	Mounting Google Drive	45
Figure 5.4	Installing and verifying YOLOv8 ultralytics library	46
Figure 5.5	YOLOv8 installation and setup check	46
Figure 5.6	Custom training and evaluating the model	47
Figure 5.7	Creation of training results and visualizations	47
Figure 5.8	YOLOv10 model setup and Roboflow dataset integration	49
Figure 5.9	Custom training and evaluating the model	49

Figure 5.10	Creation of training results and visualizations	50
Figure 5.11	YOLOv11 training tracker setup with TensorBoard.	51
Figure 5.12	YOLOv11 model setup and Roboflow dataset integration	52
Figure 5.13	Custom training and evaluating the model	52
Figure 5.14	Creation of training results and visualizations	52
Figure 6.1	Precision-Confidence Curve based on the YOLOv8 model	56
Figure 6.2	Precision-Recall Curve based on the YOLOv8 model	57
Figure 6.3	Detection for Healthy Leaves	60
Figure 6.4	Detection for Bacterial Blights	61
Figure 6.5	Detection for Brown Spots	62
Figure 6.6	Detection for Rice Blasts	63
Figure 6.7	Rice Leaf Disease Detection Dashboard	63

LIST OF ABBREVIATIONS

YOLO	-	You Only Look Once
CNN	-	Convolutional Neural Networks
IRRI	-	International Rice Research Institute
ML	-	Machine Learning
DL	-	Deep Learning
SVM	-	Support Vector Machines
KNN	-	K-Nearest Neighbor
MLP	-	multi-layer perceptron
RSW	-	RideSpider Water Wave
LOOP	-	Local Optimal Oriented Pattern
mAP	-	Mean Average Precision

LIST OF SYMBOLS

δ	-	Minimal error
r	-	Radius
Re	-	Reynold Number

LIST OF APPENDICES

APPENDIX	TITLE	PAGE
Appendix A	Gantt Chart	Error! Bookmark not defined.

CHAPTER 1

INTRODUCTION

1.1 Introduction

Rice, scientifically known as *Oryza sativa* L., is one of the staple food sources for more than half of the global population, particularly in Asia, where it is essential for economic stability and food security. Currently, over 2.7 billion people worldwide rely on rice as their primary food source, a figure that is projected to increase to 3.9 billion by the year 2025, according to the International Rice Research Institute (IRRI). This rising dependence draws attention to the crop's vital significance and highlights its supply chain's weaknesses. However, rice cultivation is continually threatened by various diseases, which can significantly reduce yield and quality. Around 15% of rice crops are wasted globally due to leaf diseases (Rathore et al., 2022). Bacterial blight, brown spot, and blast are among the most common diseases historically diagnosed through manual inspection by experts. This conventional approach is time-consuming and vulnerable to inaccuracies, highlighting the necessity of more precise and efficient diagnostic methods.

Recent advancements in digital image processing and machine learning have shown great promise in transforming agricultural practices, particularly in disease detection and management. However, accurately detecting these diseases remains challenging due to the presence of overlapping leaves and similar backgrounds in field images (Aggarwal et al., 2022). In response to these challenges, the innovative use of the You Only Look Once (YOLO) object detection algorithms represent a significant leap forward. The YOLO technology, particularly its newer iterations YOLOv8, YOLOv10, and YOLOv11, offers promising solutions with its ability to enhance the speed and accuracy of detecting and classifying rice leaf diseases. By developing a robust model that utilizes these advanced machine learning techniques, the project

seeks to provide actionable insights for early disease intervention, thus improving crop management and ensuring sustainable agricultural productivity.

1.2 Problem Background

The detection of rice leaf diseases is a significant focus in agricultural research as it has a considerable impact on crop yield and food security. Traditionally, this detection has relied on manual inspections conducted by skilled experts. While this method has been foundational in agricultural practice, it is fraught with challenges: it is labor-intensive, slow, and requires a high level of expertise, making it unsustainable at the scale needed for today's agricultural demands. Moreover, this traditional approach is susceptible to human error, leading to inconsistent and sometimes inaccurate diagnoses. To overcome these limitations, recent advances in digital agriculture have increasingly used machine learning and computer vision techniques to automate and improve the accuracy of disease detection (Sindhu et al., 2023). These technologies can potentially automate the disease detection process, enhancing its accuracy and speed.

Convolutional Neural Networks (CNNs) have become particularly effective among the various algorithms developed for image recognition because of their capacity to process and analyze images in a manner that resembles the human brain (Udayananda et al., 2022). Among the CNN-based algorithms, You Only Look Once (YOLO), a cutting-edge CNN-based algorithm, has shown exceptional promise. Yolo's architecture is uniquely suited for real-time applications as it processes the entire image in a single evaluation rather than the traditional method of scanning parts of the image sequentially (Agbulos et al., 2021). This ability enables YOLO to detect multiple objects—the diseased spots on the leaves—in one go, significantly speeding up the process and enhancing its efficiency.

However, the effectiveness of CNNs, including Yolo, heavily relies on the quality of feature selection. Feature selection involves determining the most relevant characteristics of the input data that contribute to successful predictions. For rice leaf disease detection, this means identifying specific indicators in leaf images, such as

color, texture, and morphological changes, which signal the presence of disease. Traditional feature selection methods have often needed help to handle the complexity and variability inherent in natural images, which can result in the development of less accurate predictive models (Tian et al., 2021). Given these challenges, this research aims to refine feature selection methodologies within the YOLO framework to enhance its application in detecting rice leaf diseases. This research investigates the potential to make a substantial contribution to the field of precision agriculture by enhancing the ability of YOLO to identify disease-specific features in rice leaves more accurately. This will enable the development of more effective disease management strategies that can protect crop yields and ultimately improve global food security.

1.3 Problem Statement

The detection of rice leaf diseases is critical for ensuring high crop yields and food security, yet traditional manual inspection methods are labour-intensive, slow, and prone to inaccuracies. Despite recent advances in digital image processing and machine learning, current detection systems still need help with the complexity of natural images, particularly due to overlapping leaves and similar backgrounds. This research addresses these challenges by utilizing different versions of the YOLO algorithm framework, improving the accuracy and efficiency of detecting rice leaf diseases. Enhancing these methodologies will lead to better rice plant disease management strategies, ultimately safeguarding crop production and contributing to global food security.

1.4 Research Objectives

The objectives of the project are as follows:

- (a) To conduct analysis on the dataset to gain insights, then annotate and label the images to create a finalized training dataset.
- (b) To develop machine learning models to detect and classify the three major rice plant diseases, bacterial blight, blast, and brown spot, from rice leaf images.
- (c) To develop a dashboard that visualizes model analytics, facilitates the model selection and disease detection process, and provides essential information on rice leaf diseases for general users.

1.5 Scopes of the project

- i. Exploration and comprehensive analysis of rice plant image datasets sourced from various open data repositories from Kaggle for training and testing machine learning models.
- ii. Implementation of different versions of the YOLO algorithm, including YOLOv8, YOLOv10, and YOLOv11, for rice leaf disease detection and evaluate their performance to enhance accuracy and computational efficiency.
- iii. Development of a dashboard to visualize and compare the model analytics and provide essential information.

1.6 Expected output

The implementation of feature selection within the YOLO framework for detecting rice leaf diseases is expected to significantly improve the accuracy and efficiency of disease identification in rice plants. By utilizing advanced machine learning techniques, particularly various versions of the YOLO algorithm, this project offers a robust solution to the challenges posed by the complexity and variability of

natural images in agricultural contexts. Traditional methods of disease detection often struggle with the overlapping of leaves and similar backgrounds, leading to inaccuracies. The YOLO-based approach addresses these limitations by improving feature selection to capture better disease-specific characteristics such as colour, texture, and structural changes.

In the context of stakeholders, including farmers and agricultural experts, the proposed solution will enable more precise and timely disease management. Accurate detection and classification of rice leaf diseases contribute to more effective intervention strategies, ultimately enhancing crop health and productivity. Integrating YOLO algorithms streamlines traditional agricultural practices, offering a more automated and reliable diagnostic tool. The model's feature selection capabilities reduce false positives and negatives, ensuring that disease management is both proactive and accurate. The expected outcome of this project includes a transformative impact on rice disease detection and management. Through improved diagnostic accuracy, better disease management strategies, and enhanced crop yield, the project contributes to sustainable agricultural productivity and global food security. This advancement will provide a more efficient and transparent process for managing rice crops, benefiting the entire agricultural sector.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

This chapter discusses the detection of rice leaf diseases using the YOLO algorithm framework. This section provides a thorough analysis of the common diseases affecting rice leaves, specifically blast, brown spot, and bacterial blight. It reviews the working principles and development of the YOLO algorithm as an object detection model, particularly its application in agricultural contexts. Additionally, the stages of the detection method, including feature selection and image processing techniques, are briefly covered. This chapter also includes a comparison of various YOLO versions and other machine learning models to evaluate their performance, accuracy, and detection speed in identifying rice leaf diseases. Through this comprehensive review, the effectiveness of different approaches and technologies in improving disease detection in rice plants is analyzed and discussed.

2.2 Overview of the rice leaf diseases

Rice, a highly consumed staple crop worldwide, faces significant challenges from many diseases that impact its foliage, resulting in major reductions in productivity and quality. Gaining a comprehensive understanding of these diseases, including their symptoms and the consequences they have, is essential in order to develop efficient strategies for detecting and managing them. This section presents an in-depth review of the most common and significant rice leaf diseases: blast, brown spot, and bacterial blight (Canlas et al., 2021).

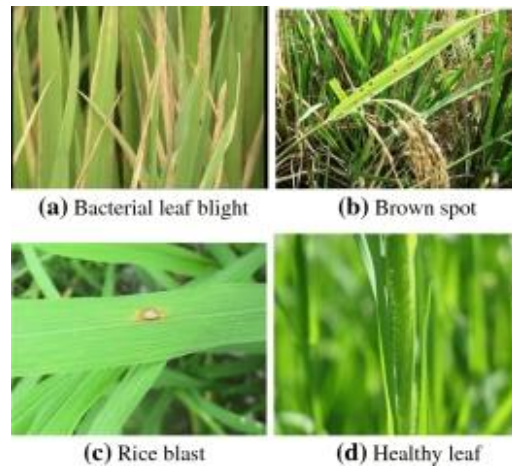


Figure 2.1 (a), (b), (c) shows leaves with diseases. (d) shows healthy leaf

2.2.1 Bacterial Blight

Bacterial blight is one of the most severe diseases that affect rice plants. The yield loss becomes greater when the disease occurs earlier. Bacterial blight can result in more than 70% yield loss when susceptible varieties are cultivated in environments that are conducive to the disease (Niones et al., 2021). Bacterial blight does not affect yield when plants are infected during the booting stage; however, it does result in a high proportion of broken kernels and poor-quality grains.

The pathogen causing the bacterial blight in rice is *Xanthomonas oryzae* pv. *oryzae*, a bacterium that causes major harm to the crop. The disease primarily impacts the plant's vascular system, resulting in wilting and death. The disease shows initially as water-soaked streaks on the leaf blades and sheaths, which then get larger and become yellow, resulting in the formation of large, uneven lesions (Sanya et al., 2022). These lesions can merge, resulting in the coverage of extensive areas of the leaf and leading to the premature drying out and death of the leaf. It can manifest in both tropical and temperate habitats, particularly in irrigated and rainfed lowland regions. Typically, the disease develops in temperatures ranging from 25 to 34°C, along with a relative humidity level of 70%.



Figure 2.2 Bacterial blight disease on rice leaf

2.2.2 Brown Spot

The brown spot has been recognized as one of the most common and damaging rice plant diseases. Brown spot leads to both quantity and quality losses. The disease leads to an average reduction in yield of 5% in all lowland rice production in South and Southeast Asia, and a highly infected field might suffer a production loss of up to 45%. Infected seeds produce seedling blight, resulting in 10-58% mortality. It also has an impact on the quality and quantity of grains per panicle, as well as kernel weight.

Bipolaris oryzae (formerly *Helminthosporium oryzae*) is the fungus that causes brown spot, which affects rice leaves, kernels, and seedlings. The disease can be identified by the presence of small, circular to oval brown lesions with a distinct dark brown margin and a grey or light brown centre (Upadhyay & Kumar, 2021). These spots can coalesce to form larger patches, leading to extensive leaf damage. Brown spot lesions in certain varieties of rice may be mistaken for blast lesions. To confirm, look for circular, brownish dots with a grey centre surrounded by a reddish edge. According to the International Rice Research Institute (IRRI), the disease can develop in areas with high relative humidity (86–100%) and temperatures between 16 and 36°C, commonly occurring in unflooded, nutrient-deficient soils or soils that

accumulate toxic substances. Infection requires leaves to be wet for 8–24 hours. The fungus can survive in seeds for over four years and spread between plants through the air.



Figure 2.3 Brown spot disease on rice leaf

2.2.3 Rice Blast

Rice blast, caused by the fungus *Magnaporthe oryzae*, is a severely destructive rice disease that has a global impact. The threat of rice blast is a significant challenge in addressing global food insecurity since this disease is the cause of around 30% of worldwide rice production losses. This loss is equivalent to the food supply for 60 million individuals (Nalley et al., 2016). This disease can impact several aerial components of a rice plant, including the leaf, collar, node, neck, certain areas of the panicle, and sometimes the leaf sheath (Asibi et al., 2019). During later growth phases, a severe leaf blast infection leads to a decrease in leaf area available for grain fill, reducing grain yield. Rice plants at the seedling stage can be fatally affected by leaf blasts, resulting in significant yield losses in instances of severe infection. A leaf blast infection can be fatal for seedlings or plants until they reach the tillering stage.

Blast can occur wherever blast spores are present. It is found in regions with low soil moisture, frequent and long-term rain showers, and cool temperatures during the day. The disease is more likely to develop in upland rice due to the large day-night temperature differences that induce dew formation on leaves and the generally lower temperatures (Neupane & Bhusal, 2021). Blasts can occur during all phases of rice development. According to the International Rice Research Institute (IRRI), initial symptoms of rice blast appear as white to grey-green lesions or spots with dark green borders. As the disease progresses, older lesions on the leaves become elliptical or spindle-shaped with whitish-to-grey centres and red-to-brownish necrotic borders. Blast lesions are frequently mistaken for brown spot lesions. Leaf blast lesions usually show an elongated shape with pointed ends, whereas brown spot lesions generally seem spherical, brown in colour and have a yellow halo surrounding the lesion.

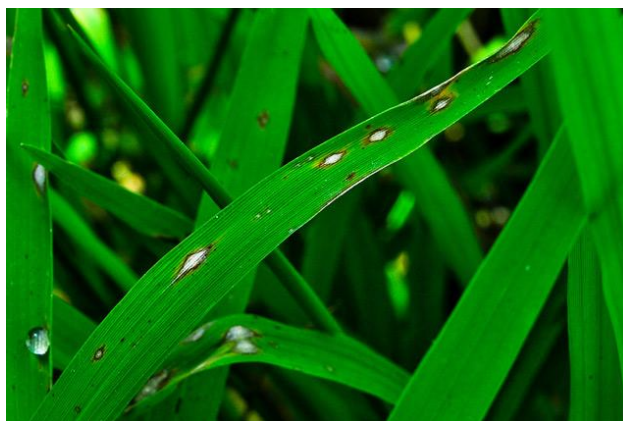


Figure 2.4 Blasts disease on rice leaf

2.3 Overall Impact and Challenges

The combined impacts of these diseases pose a substantial risk to global rice production, affecting both the availability of food and the stability of economies, particularly in places that heavily rely on rice as a primary source of livelihood. Every disease poses specific challenges when it comes to identifying, handling, and controlling it. Traditional disease identification methods, which heavily rely on visual inspections conducted by professionals, are not only demanding in terms of labour and

time but also prone to errors, particularly during the initial phases of infection or in cases of the simultaneous presence of multiple diseases (John et al., 2023).

The variability in symptoms and environmental conditions further complicates disease management. To achieve effective disease prevention and management, it is necessary to use an integrated approach that combines the use of resistant cultivars, cultural practices, and, when suitable, chemical control methods. However, the process of creating and implementing resistant cultivars can be time-consuming, and the efficacy of chemical controls is sometimes constrained by factors such as expense, accessibility, and ecological considerations. The progress in digital image processing and machine learning, particularly the implementation of YOLO algorithms for disease identification, presents encouraging new resources to improve the accuracy and effectiveness of disease control in the cultivation of rice.

2.4 Existing Rice Diseases Detection Method

The presence of rice diseases poses a substantial risk to rice production worldwide, requiring the implementation of efficient methods for detection and control. With time, the techniques used to identify rice diseases have progressed from conventional methods to advanced technologies, driven by the necessity for enhanced precision, effectiveness, and scalability. This section presents a brief overview of current methods for detecting diseases in rice, ranging from conventional techniques to modern advancements.

2.4.1 Traditional methods of disease detection

Traditional methods for detecting diseases in rice plants include expert visual inspection, diagnosis based on symptoms, laboratory testing, use of disease identification guides and manuals, and reliance on farmer knowledge and experience.

Visual inspection includes specialists examining plants to identify visible symptoms. However, this method is time-consuming, labour-intensive, and limited by the number of available experts. The process of diagnosing based on symptoms relies on the use of manuals and visual guides. However, it faces challenges due to the similarity of symptoms, difficulties detecting early-stage symptoms, and the influence of environmental factors (Kaur, Guleria, & Trivedi, 2021). Laboratory testing provides accurate outcomes but requires significant resources, consumes a considerable amount of time, and is frequently unavailable in rural areas. Disease identification guides provide valuable information, although they may not address all variations of symptoms or newly emerging diseases, and their efficacy relies on the user's interpretation. Finally, the knowledge and expertise of farmers are valuable, but their accuracy and consistency can vary considerably. Moreover, farmers in rural areas may not have access to up-to-date information on diseases and effective management techniques.

2.4.2 Advanced methods of disease detection

The integration of advanced techniques, specifically machine learning (ML) and deep learning (DL), has revolutionized the field of rice disease detection. These techniques have significantly enhanced accuracy, efficiency, and scalability compared to traditional methods. Machine learning algorithms, such as Support Vector Machines (SVM), Decision Trees, Random Forests, and K-Nearest Neighbours (KNN), are used to analyze extensive datasets of plant images in order to detect and categorize disease symptoms (Ahmed et al., 2019). These algorithms show a high level of accuracy, but they face difficulties such as parameter optimization and computational complexity. Deep learning techniques, particularly Convolutional Neural Networks (CNNs) and You Only Look Once (YOLO), are highly effective in identifying and categorizing rice diseases in images (Kiratiratanapruk et al., 2020). However, these techniques necessitate extensive datasets and substantial computational resources. Further, hybrid and ensemble methods, which involve combining multiple machine learning (ML) and deep learning (DL) techniques, can improve the resilience and precision of models by

capitalizing on the unique capabilities of various algorithms. Although these advanced methods have benefits, they encounter difficulties concerning data quality, computational resources, the ability to apply the model in different scenarios, and the requirement to combine them with conventional knowledge to ensure practical effectiveness.

2.5 Concept of YOLO

The You Only Look Once (YOLO) algorithm marks an important step forward in the field of real-time object detection. Developed by Joseph Redmon and his team in 2015, YOLO revolutionized the approach to object detection by enabling models to analyze images and identify objects in a single evaluation (Vijayakumar & Vairavasundaram, 2024). Compared to conventional multi-stage detection methods, YOLO utilizes a single convolutional neural network (CNN) to predict bounding boxes and class probabilities directly from full images. The YOLO algorithm works by dividing the input image into a grid of $S \times S$ cells. Each cell in the grid is responsible for predicting a specific number of bounding boxes and their respective class probabilities (Diwan, Ani, & Tembhurne, 2022). Each bounding box is defined by its coordinates, width, height, and confidence score, indicating the accuracy level in detecting an object. YOLO uses advanced techniques like Non-Maximal Suppression (NMS) to remove redundant bounding boxes, keeping only the most precise ones (Redmon et al., 2015). This single-pass detection approach, greatly enhances its speed and efficiency. By processing the entire image in one go, YOLO achieves remarkable real-time performance, making it suitable for various applications where speed and accuracy are important.

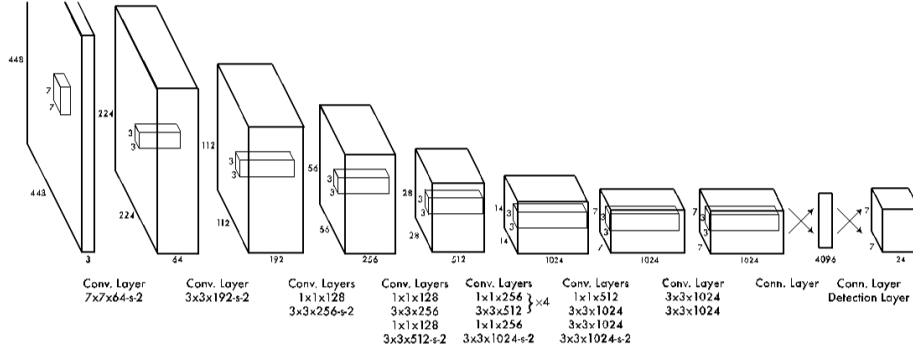


Figure 2.5 Model architecture of YOLO (Redmon et al., 2015)

2.5.1 Evaluation of YOLO

The YOLO algorithm has undergone many iterations since its development, with each iteration enhancing the previous versions. YOLOv1, which was introduced in 2015, marked the start of real-time object detection with reasonable accuracy. However, it faced challenges in terms of low detection rates and significant localization errors (Kim et al., 2021). YOLOv2 improved upon these challenges by integrating batch normalization and higher-resolution classifiers, resulting in a noticeable enhancement in performance (Gupta & Devi, 2020). YOLOv3 further improved the architecture with modern CNNs, enhancing the detection of small targets and overall accuracy. YOLOv4 implemented advanced techniques to enhance image augmentation, and introduced a new CNN architecture called CSPNet to further improve feature extraction and network efficiency (Viswanatha, Chandana, & Ramachandra, 2022). YOLOv5, built on a flexible and lightweight framework, offers similar performance to YOLOv4 while being more convenient to train and deploy. Subsequent versions like YOLOv7, YOLOv8, and the latest YOLOv9, have integrated features such as spatial pyramid pooling, anchor-free mechanisms, and advanced transformer-based architectures, pushing the limits of real-time object detection with much more improved precision and efficiency. These enhancements make YOLO a powerful tool for various applications, including detecting rice leaf diseases, where high accuracy and real-time processing are critical.

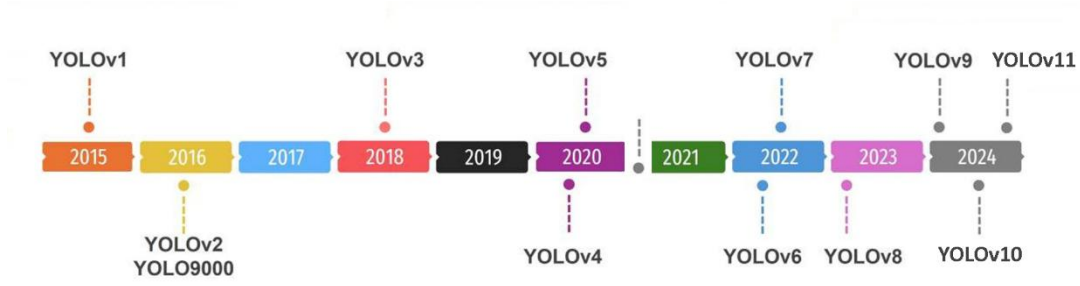


Figure 2.6 Evaluation of YOLO versions

2.5.2 YOLO v8

A further developed version of the YOLO architecture, YOLO v8, was released in 2023. It integrates techniques to expand the boundaries of object detection in terms of both speed and precision. The backbone still utilizes an optimized version of the CSPDarknet, which further enhances its ability to extract features. The neck is improved with additional layers for enhanced feature aggregation, building upon the PANet structure introduced in YOLOv5. Instead of using the conventional anchor-based method, YOLOv8 uses an anchor-free approach to enhance performance (Bakirci, 2024). The integration of enhanced feature pyramid networks in YOLOv8 improves the ability to handle objects of different sizes, resulting in higher accuracy rates (Yang et al., 2023). Moreover, YOLOv8 uses advanced data augmentation techniques, such as MixUp and CutMix, to improve the model's robustness and generalization (Zhang et al., 2024).

2.5.3 YOLO v10

YOLOv10, released on May 23, 2024, enhances real-time object detection by overcoming previous limitations, like dependence on Non-Maximum Suppression (NMS) and architectural inefficiencies. The approach implements consistent dual assignments for NMS-free training and adopts a comprehensive model design driven by efficiency and accuracy (Wang et al., 2024). The architecture improves feature extraction process through an enhanced CSPNet backbone, uses PAN layers in the neck for efficient multi-scale feature fusion, and utilizes both one-to-many and one-

to-one heads to deliver full supervisory signals during training while eliminating NMS during inference, thus minimising latency. The main features comprise lightweight classification heads, spatial-channel decoupled down sampling, rank-guided block design, large-kernel convolutions, and partial self-attention modules, which together improve performance while maintaining low computational costs. YOLOv10 utilizes dual-label assignments and extensive model optimisations to achieve enhanced precision and efficiency, rendering it suitable for various applications (Alif & Hussain, 2024).

2.5.4 YOLO v11

YOLOv11, released on September 30th, 2024, represents the most recent advancement in the Ultralytics YOLO series of real-time object detectors, enhancing accuracy, speed, and efficiency in computer vision. YOLOv11 enhances prior versions through notable architectural and training improvements, defining it as a versatile option for various tasks. The key features consist of an enhanced backbone and neck architecture that improves feature extraction for more accurate object detection, as well as refined architectural designs and optimized training pipelines that provide increased processing speeds while ensuring a balance between accuracy and performance. YOLOv11 introduces significant architectural improvements, including the C3k2 (Cross Stage Partial with kernel size 2) block, SPPF (Spatial Pyramid Pooling - Fast), and C2PSA (Convolutional block with Parallel Spatial Attention) components, which collectively enhance feature extraction and improve overall model performance (Khanam & Hussain, 2024). The system's adaptability across diverse environments, like the latest devices, cloud platforms, and NVIDIA GPU-supported systems, ensures optimal deployment flexibility. YOLOv11 allows various computer vision tasks, such as object detection, instance segmentation, image classification, pose estimation, and orientated object detection (OBB).

2.5.5 Application of YOLO in Crop Disease Detection:

The YOLO algorithm has grown into an effective technique for detecting objects in real-time, making it particularly well-suited for agricultural purposes, such as identifying crop diseases. YOLO's rapid and precise object recognition capabilities have important implications for enhancing crop health management and productivity (Morbekar, Parihar, & Jadhav, 2020). The implementation process entails developing of a well-arranged dataset by gathering and annotating images of crops affected by different diseases. This dataset is then used to train the YOLO model, implementing techniques like transfer learning and data augmentation. Finally, the model is deployed for real-time inference on platforms such as drones, mobile applications, and IoT devices. This facilitates the effective and prompt identification of diseases, which is essential for managing wide agricultural regions.

YOLO algorithm provides numerous advantages for disease detection in agriculture, such as speed and efficacy, high precision, the ability to scale up, and automation. Due to its ability to process images in a single iteration, this algorithm is one of the most rapid object detection algorithms currently accessible, which is essential for applications that require real-time processing. When YOLO is trained correctly, it can achieve a high level of accuracy in detecting and classifying diseases. It is also capable of thoroughly analyzing multiple objects in an image. YOLO, which is scalable to large agricultural areas, can be combined with drones and IoT devices for ongoing monitoring (Chen et al., 2023). In addition, the automation of disease detection decreases the need for labour-intensive manual inspections, resulting in accurate and consistent diagnostics that improve disease management strategies.

2.6 Enhanced Feature Selection

The selection of the polygon technique for annotating rice leaf diseases in Roboflow significantly enhances the training process for disease detection models.

Traditional bounding boxes are simple and easy to apply but often fail to accurately capture the irregular and intricate patterns that rice leaf diseases exhibit, leading to reduced localization precision. Polygon labeling, utilizing advanced tools like Meta AI's Segment Anything Model (SAM) and Roboflow's Smart Polygon feature, helps annotators define the impacted regions more accurately, capturing the true shape and scale of disease symptoms (Roboflow, 2023). This precise annotation ensures that the model learns to identify the distinct features of different diseases, such as spots, blasts, and blights, resulting in enhanced precision and consistency in detection. Polygon labeling improves the precision of training data by capturing exact boundaries, essential for machine learning models to understand the intricacies of disease manifestation. This leads to better detection accuracy, particularly for complex objects, and enhances the model's diagnostic capabilities in real-world agricultural scenarios (Zhu, Hao, & Ma, 2023). Experiments have shown that models trained with polygon-labeled data outperform those using bounding boxes, especially when augmented with techniques like rotation and cutout, enhancing robustness and generalization (Roboflow, 2023). This approach boosts object detection accuracy and reliability, benefiting applications in autonomous driving, robotics, and surveillance.

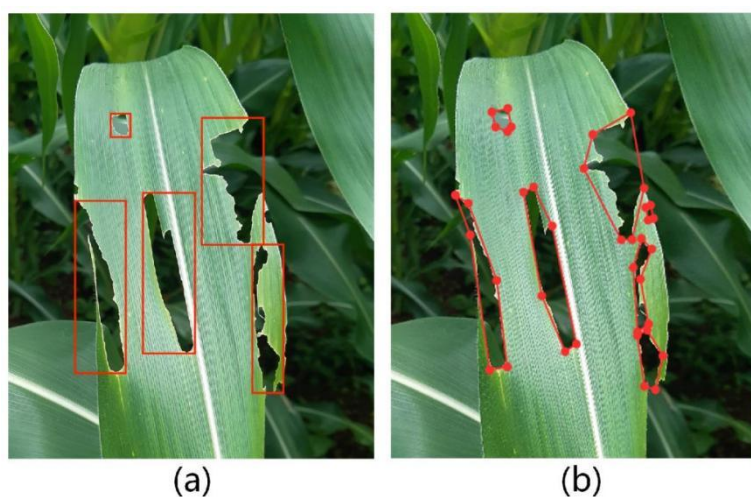


Figure 2.7 Part (a) shows traditional object annotations, and part (b) shows polygon object annotations

2.7 Related works

In the research paper "Alpha-EIOU-YOLOv8: An Improved Algorithm for Rice Leaf Disease Detection," Trinh et al. (2024) presented a new method aiming at improving the detection and identification of rice leaf diseases. The research utilized a modified YOLOv8 algorithm, substituting the conventional Box Loss function with a hybrid of EIoU and α -IoU loss to enhance performance. A comparative analysis of YOLOv7 and YOLOv5 models revealed that the proposed YOLOv8 approach attained the highest accuracy at 89.9%. The researchers provided a dual-phase process. Initially, images of rice leaf diseases were systematically gathered and classified into three categories: blast leaf, leaf folder, and brown spot. In the second stage, the YOLOv8 model underwent training on a newly created dataset comprising 3175 images, with 2608 allocated for training, 326 for validation, and 241 for testing.

In the study "Deep Learning Based Model for Rice Plant Disease Classification Using YOLOv8," Ismail et al. (2023) proposed a method to identify rice plant diseases using the YOLOv8 algorithm. The researchers collected 600 images of healthy and unhealthy rice plants captured using Android smartphones under a white background. These images were used to train the YOLOv8 model, which was implemented in Google Colab with access to faster GPUs and TPUs. The model underwent data augmentation techniques to improve its performance. The study reported a classification accuracy of 90% for both training and validation after 50 epochs, demonstrating the model's effectiveness in binary classification of rice plant diseases. The findings emphasize the potential of YOLOv8 in enhancing the accuracy and efficiency of disease detection in agricultural settings.

In the research study "Enhanced YOLOv8 Algorithm for Leaf Disease Detection with Lightweight GOCR-ELAN Module and Loss Function: WSIoU," Wen et al. (2025) presented a modified technique for rice leaf disease detection. To tackle issues such as changing target sizes, occlusions, and detection errors in complex surroundings, this method improves the YOLOv8 network. By changing the C2f modules in the Backbone with the GOCR-ELAN module, the suggested approach

lowers parameters and enhances feature extraction. To further reduce the complexity of the model and better handle occlusion concerns, the ADown module takes the role of CBS convolution. The paper presents the WSIoU loss function for improved convergence and localisation precision in order to decrease errors and increase detection accuracy. According to experimental results, the model increases MAP50 from 86% to 87.7% while reducing parameters by 28.7% and GFLOPs by 43.2%.

The research paper titled "Early Disease Diagnosis for Rice Crop" by Masood et al. (2020) introduced an innovative method for detecting rice leaf diseases using Mask R-CNN. This model combines localized classification and segmentation to offer detailed information about diseased regions on rice plants. The model achieves an overall accuracy of 87.6% using a dataset with detailed annotations for each infected segment, significantly outperforming traditional CNN methods. Mask R-CNN predicts the category, bounding box, and mask for every affected area, providing precise and early detection abilities. This study sets a new benchmark for identifying rice diseases, highlighting the importance of localized data in enhancing prediction accuracy and offering practical solutions for real-time field use.

Table 2.1 Summary of related works

Author	Title	Methods	Outcome	Limitations
(Trinh et al., 2024)	Alpha-EIOU-YOLOv8: An Improved Algorithm for Rice Leaf Disease Detection	Modified YOLOv8 algorithm	Achieved accuracy of 89.9%	Lack of comparison with newer YOLO models, limits a thorough assessment of the proposed method compared to the recent advances in object detection.
(Ismail et al., 2023)	Deep Learning Based Model for Rice Plant Disease Classification Using YOLOv8.	YOLOv8 algorithm	Accuracy of 90% is achieved	The model was limited to binary classification. Reliance on high-quality images and trained on small dataset.
(Haque et al., 2022)	Rice Leaf Disease Classification and Detection Using YOLOv5	YOLOv5, Baseline CNN	Achieved accuracy of 90% and F1-score of 81%	Challenges in data collection, pre-processing, and handling varying disease types.
(Wen et al., 2025)	Enhanced YOLOv8	YOLOv8 with GOCR-ELAN	Achieves an accuracy of	High computational requirements, difficulty

	algorithm for leaf disease detection with lightweight GOCR-ELAN module and loss function: WSIoU		87.7%,	generalizing across various rice types and conditions, and reliance on high-quality images.
(Masood et al., 2020)	Early Disease Diagnosis for Rice Crop	Mask R-CNN, Baseline CNN	Mask R-CNN: Accuracy 87.6% F1-score 88.6% Baseline CNN: Accuracy of 58.4%, F1-score of 62.8%	Challenges in image preprocessing and handling variable lighting conditions.

CHAPTER 3

RESEARCH METHODOLOGY

3.1 Introduction

This chapter presents an in-depth description of the research framework that corresponds to the research objectives. The framework provides a detailed explanation of each phase, guided by the flowchart as illustrated in Figure 3.1. The five phases are: 1. Problem Finding and Planning, 2. Data Exploration, Annotation and Dataset Finalization 3. Model Development and Training 4. Model Testing and Performance Evaluation and 5. Dashboard Development. This chapter also includes the software and hardware requirements for the project. Each phase is designed to systematically address the research objectives and ensure the development of accurate and efficient YOLO-based models for detecting rice leaf diseases.

3.2 Project Framework

The project framework is important as it provides a structured and systematic approach to achieving the research objectives. Each phase within the framework serves a specific purpose, ensuring that the research progresses in an organized and efficient manner. The project framework for the research is given in the following:

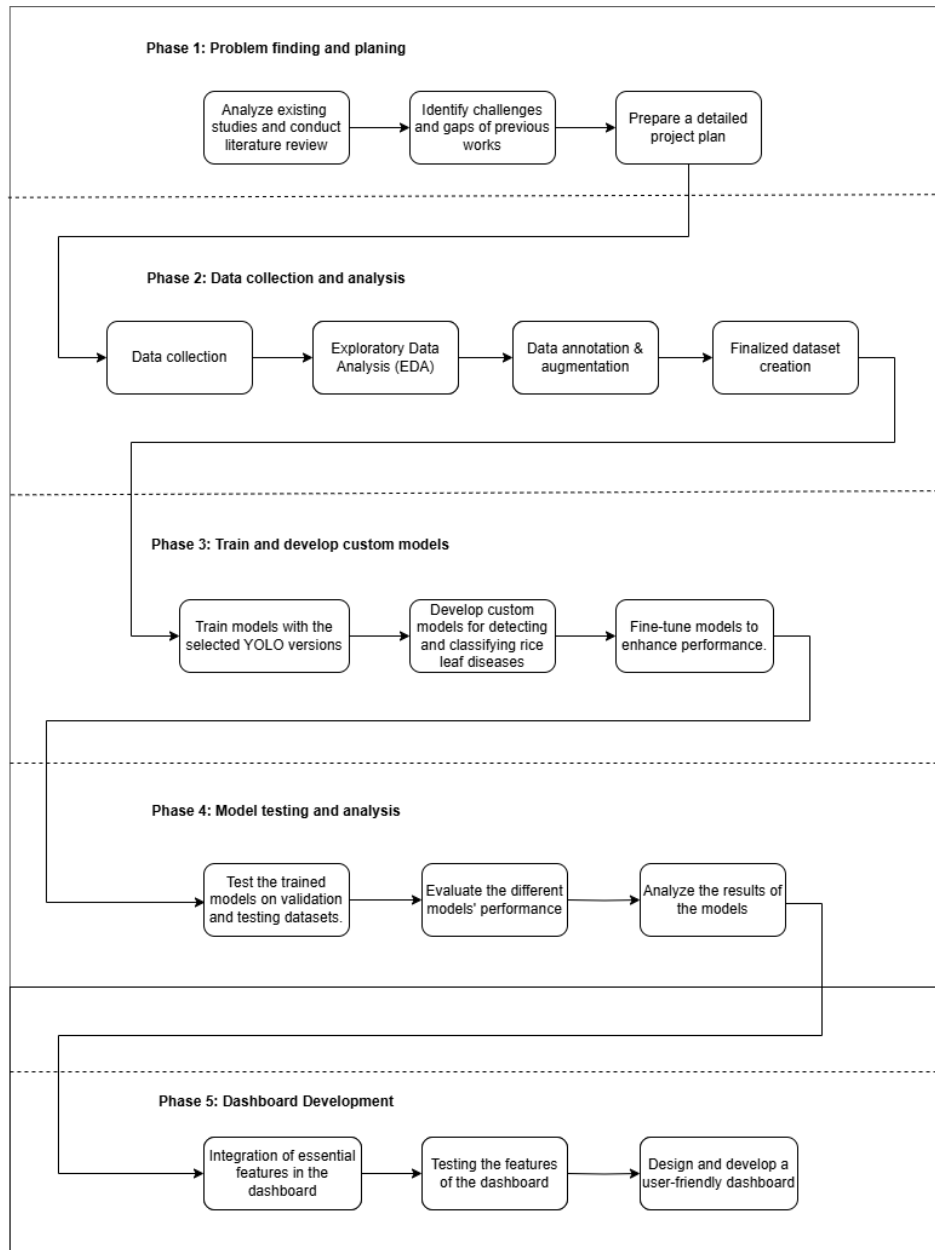


Figure 3.1 Project Framework

3.2.1 Phase 1: Problem Finding and Planning

In Phase 1, the initial step is to conduct a thorough literature review to review existing research and current studies related to the detection of rice leaf diseases. This review includes an analysis of different approaches utilized for the detection of rice leaf diseases, with a particular emphasis on evaluating the strengths and drawbacks of both traditional and modern techniques. Key areas of investigation include the effectiveness of machine learning and deep learning algorithms, with a specific focus

on the YOLO (You Only Look Once) framework. By understanding the current state of research, the study identifies gaps, challenges, and opportunities for improvement in the detection of rice leaf diseases.

Based on the existing literature review, this step involves a critical analysis of the identified problems in previous work. It includes understanding the limitations of traditional manual inspection methods, such as labor intensity, time consumption, and susceptibility to human error. Furthermore, the review of modern techniques reveals issues related to the complexity and variability of natural images, particularly with overlapping leaves and similar backgrounds. This analysis helps identify precise areas for potential improvements, such as enhancing feature selection methodologies to enhance both accuracy and efficiency.

The final step in Phase 1 is to prepare a comprehensive project plan based on the insights gained from the literature review and problem analysis. This plan outlines the objectives, scope, and methodology of the research. It details the phases of the project, including data collection and annotation, model training and development, model testing and analysis, and lastly the development of the dashboard. The project plan also includes timelines, resource allocation, and objectives to ensure the research progresses in a structured and systematic manner. By preparing a detailed project plan, the research is set on a clear path towards developing robust models for detecting rice leaf diseases using the YOLO algorithm framework.

3.2.2 Phase 2: Data Collection and Analysis

Data Collection:

The first step in Phase 2 involves gathering image datasets of rice leaves affected by various diseases. The datasets are obtained from different open data repositories on Kaggle, which offer a diverse range of rice leaf images. The focus is on obtaining high-quality images that represent different types of rice leaf diseases, primarily bacterial blight, brown spot, and blast. Ensuring a diverse and

comprehensive dataset is crucial for developing a robust detection model capable of generalizing well across different environmental conditions.

Table 3.1 Details of the datasets

Dataset Name	Size	File Type	Number of Healthy Leaf Images	Number of Bacterial Blight Images	Number of Brown Spot Images	Number of Leaf Blast Images
Rice Leaf Disease DataSet	2.2 GB	jpg	749	829	1103	1252
Rice Leaf Diseases Detection	4.17 GB	jpg, png	2967	2959	3406	3911

Data Verification:

To ensure the reliability and accuracy of the data used in this research, several verification steps were implemented. The datasets were sourced from credible open repositories on Kaggle, which were validated for proper documentation, citations, and user reviews. A thorough consistency check was conducted to ensure uniformity in image formats, naming conventions, and resolutions. Duplicate images and annotations were identified and removed using automated scripts to eliminate redundancy and maintain the integrity of the dataset. Additionally, the annotated datasets underwent a manual review to ensure the polygon annotations precisely outlined the diseased regions on rice leaves, avoiding mislabeled or incomplete annotations.

Class distribution across the dataset was analyzed to ensure balanced representation of all categories, including Brown Spot, Healthy Leaf, Bacterial Blight, and Rice Blast. Data augmentation techniques such as rotation, scaling, and flipping were applied to address any class imbalances and enhance the robustness of the training process. Environmental noise, including overlapping leaves, inconsistent lighting conditions, and background clutter, was inspected, and unsuitable images were excluded from the dataset. By documenting all preprocessing steps, annotation

processes, and dataset manipulations, this research ensures transparency and reproducibility, thereby reinforcing the credibility and reliability of the results.

Polygonal Image Annotation:

The image annotation process was meticulously carried out using Roboflow to prepare high-quality training data for the YOLO models. The polygon annotation method was used to precisely outline diseased regions on rice leaves, including bacterial blight, brown spots, and rice blasts. Annotators began by importing the collected dataset into Roboflow, which automatically organized the images into categories for easier management. Using the Polygon Tool in Roboflow, annotators initiated the process by clicking on a starting point on the diseased area of an image and adding successive points to trace the shape of the affected region. This process allowed for accurate labeling of irregular disease patterns, ensuring the annotations conformed closely to the edges of the lesions or spots while avoiding extraneous background information.

To refine the annotations further, annotators were able to add, edit, or move vertices after the polygon was closed, ensuring the shapes matched the disease boundaries as accurately as possible. Advanced features such as auto-suggestion tools and Roboflow's Smart Polygon Tool were used to accelerate the process by providing recommendations for vertices placement based on the image's features. Each annotated region was then tagged with one of four predefined classes: Healthy Leaf, Bacterial Blight, Brown Spot and Rice Blast. Once the annotations were completed, the dataset underwent a thorough quality assurance review to ensure consistency and precision. This detailed annotation process using Roboflow provided high-quality labeled data, significantly enhancing the training effectiveness of the YOLO models and improving their ability to detect and classify rice leaf diseases with greater accuracy.

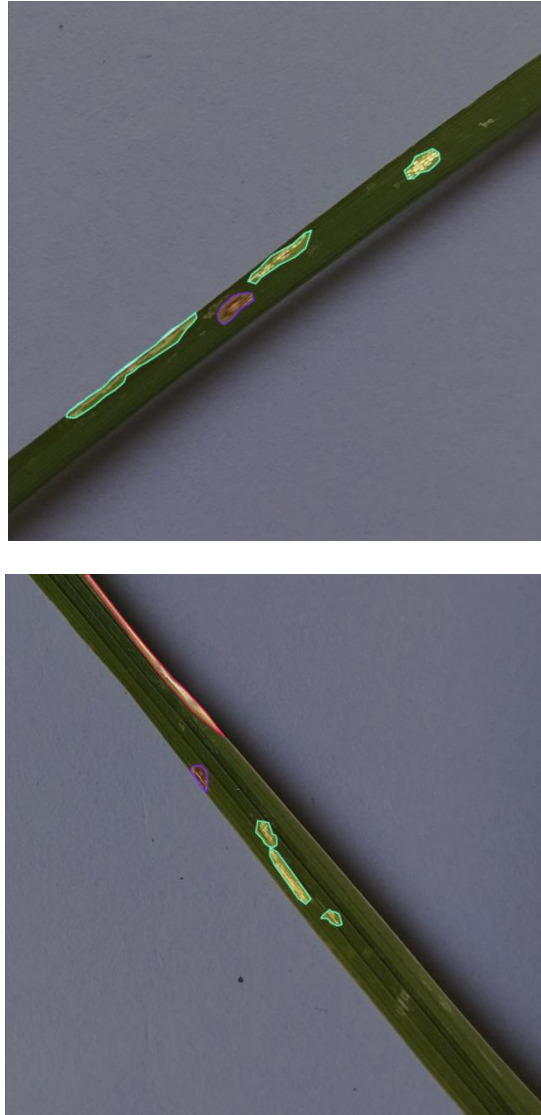


Figure 3.2 Polygonal annotation and labelling of the images of Roboflow

Data Augmentation:

Lastly, to further improve the robustness and performance of the model, data augmentation techniques are applied to the labelled dataset. These techniques include operations such as rotation, flipping, scaling, and colour adjustments, which increase the diversity of the training data. Augmentation helps in simulating various real-world conditions, making the model more resilient to changes in lighting, orientation, and other environmental factors. By expanding the dataset through augmentation, the chances of overfitting are reduced, and the model's generalization capability is enhanced, leading to better performance in real-world applications.

3.2.3 Phase 3: Train and Develop Custom Models

The initial step of Phase 3 is to finalize the dataset to train the model. This step includes organizing the collected and augmented data into distinct sets for training, validation, and testing. The training set is used to train the models, the validation set is used to tune hyper parameters and prevent overfitting, and the test set is reserved for evaluating the final model performance. Ensuring a balanced and appropriate dataset is important for developing robust models that can generalize well to new, unseen data.

Next, the datasets will be used to train models with the selected YOLO algorithm versions, specifically YOLOv8, YOLOv10, and YOLOv11, each of which provides unique architectures and improvements. Experimenting with multiple YOLO versions helps determine the best fit for rice leaf disease detection. During the training process, the model parameters are configured to meet the specific requirements of rice leaf disease detection, including the learning rate, batch size, network architecture, and other hyperparameters. Utilizing powerful computational resources like GPUs, the training process enables the models to efficiently learn to identify and classify various types of rice leaf diseases based on the annotated images.

After training the models, their performance is thoroughly evaluated to determine the most effective version. This evaluation includes assessing models using various metrics such as accuracy, precision, recall, F1 score, and mean Average Precision (mAP), providing a comprehensive understanding of each model's capability in detecting rice leaf diseases. A comparative analysis is conducted to identify which YOLO version offers the best balance of accuracy and computational efficiency, ensuring the selected model is suitable for real-time detection.

3.2.4 Phase 4: Model Testing and Analysis

In Phase 4, the first step is to thoroughly test the trained YOLO models using test datasets to ensure their effectiveness. This test dataset comprises images that were not used during the training or validation phases, providing an unbiased evaluation of

the model's performance. The goal is to evaluate how well the models can detect and classify rice leaf diseases in new, unseen data. The testing procedure involves feeding the test data into the trained YOLO models and recording their predictions, evaluating the models' ability to accurately identify and classify different types of rice leaf diseases based on the enhanced feature selection techniques applied during training. Performance metrics such as accuracy, precision, recall, F1 score, and mean Average Precision (mAP) are used to quantify the models' performance, helping to understand their strengths and weaknesses.

Once the models have been tested, the next step is to visualize the results to interpret and present the model's performance effectively. Visualization is crucial for understanding how the models make predictions and for identifying any patterns or anomalies in their performance. Using visual tools such as confusion matrices, precision-recall curves, and IoU maps helps display the model's performance, making it easier to interpret the results and understand the model's effectiveness in detecting rice leaf diseases. By visualizing the results, the strengths and areas for improvement of each model are highlighted, identifying specific instances where the models perform well and where they may fail, providing insights into potential refinements. Comparative visualizations between different YOLO versions are essential for showcasing which model performs best under various conditions, helping make informed decisions about the most suitable model.

3.2.5 Phase 5: Dashboard Development

The dashboard was developed using Streamlit, simplifying the development of a simple and interactive tool for analyzing rice leaf diseases. By utilizing simple streamlit features the sidebar for navigation, allowing user transitions between screens such as disease detection, model comparison, and dataset analytics. It also made it easy to add features like image upload by integrating the models, where users can upload rice leaf images and get detection results using YOLO models. I used Streamlit's built-in tools to create charts, tables, and other visuals to show model performance and comparisons clearly. With Streamlit, the python codes were directly utilized to handle user inputs, display results, and even customize the dashboard's

layout and design. This made it simple to create a dashboard that is interactive, looks modern, and works well for people with different levels of technical experience.

3.2.6 Algorithm Setup

The algorithm configuration for model training includes a few important steps and components to ensure the optimal performance of the YOLO models. The initial step is to develop the model architecture, which outlines the structure and configuration of the neural network. This architecture consists of convolutional layers followed by fully connected layers, with the number and size of these layers varying based on the specific YOLO variant used. For this project, YOLOv8, YOLOv10, and YOLOv11 are selected for model training due to their advancements in accuracy and computational efficiency.

The development environment for the algorithm is set up in Google Colab using Python. Necessary libraries are installed using the pip command, with PyTorch and Ultralytics serving as the primary machine learning frameworks for developing deep learning models. These frameworks are particularly effective for applications like object recognition. The YOLO library package has been optimized for efficient computation and memory management, leveraging hardware accelerators such as GPUs to facilitate faster training.

To incorporate YOLO, the git clone command is used to download the relevant repositories directly from GitHub into Google Colab. This project specifically clones the repositories for YOLOv8, YOLOv10, and YOLOv11. Following the download, a model summary is generated, providing a comprehensive overview of the model's layers, shapes, and parameters, which aids in model selection and comparison. Essential dependencies, including torch, pyplot, numpy, and cv2, are imported concurrently to provide the necessary functionality, tools, and resources required for the training process. This setup ensures a robust foundation for developing and refining the YOLO models to detect rice leaf diseases accurately.

3.2.7 Software and Hardware Requirements

The table 3.2 given summarizes the essential software, programming language, technology, and hardware requirements for the project.

Table 3.2 Software and hardware requirements

Software	Google Colab, Jupyter Notebook-Lab, Roboflow
Programming Language	Python
Technology / Technique	Deep Learning (YOLO v8, YOLOv10, YOLOv11)
Hardware	Asus Rog Strix (Intel Core i9-13980HX Windows 11, RTX4050 6GB GDDR6, DDR5 32GB RAM) with integrated camera

3.3 Chapter Summary

This chapter gives an overview of the project methodology, which consists of four key phases: Problem Finding and Planning, Data Collection and Labeling, Training and Developing Custom Models, and Model Testing and Analysis. Each phase is systematically planned to address specific objectives, ensuring the development of accurate and efficient YOLO-based models for detecting rice leaf diseases. The methodology includes the setup and training of various YOLO algorithm versions (YOLOv8, YOLOv10, YOLOv11), focusing on optimizing feature selection and improving model accuracy. The chapter also details the software and hardware requirements used in the project.

CHAPTER 4

EXPLORATORY DATA ANALYSIS

4.1 Introduction

This chapter presents the results and evaluates the performance of the prediction models based on the YOLO algorithm in training, validation, and testing datasets. The performance of the YOLO-based models was evaluated using a range of metrics, which include precision, recall, loss, and confusion matrix. Besides that, this chapter reviews the prediction model's performance to provide an in-depth review of its detection capabilities and limitations. Lastly, the real-time detection results of the detection process will be discussed. The outcomes of this research can greatly contribute to the development of robust rice leaf disease detection.

4.2 Data Collection:

The first step in Phase 2 involves gathering image datasets of rice leaves affected by various diseases. The datasets are obtained from different open data repositories on Kaggle, which offer a diverse range of rice leaf images. The focus is on obtaining high-quality images that represent different types of rice leaf diseases, primarily bacterial blight, brown spot, and blast. Ensuring a diverse and comprehensive dataset is crucial for developing a robust detection model capable of generalizing well across different environmental conditions.

Table 4.1 Details of the datasets

Dataset Name	Size	File Type	Number of Healthy Leaf Images	Number of Bacterial Blight Images	Number of Brown Spot Images	Number of Leaf Blast Images
Rice Leaf Disease DataSet	2.2 GB	jpg	749	829	1103	1252
Rice Leaf Diseases Detection	4.17 GB	jpg, png	2967	2959	3406	3911

4.3 Exploratory Data Analysis:

In this study, Exploratory Data Analysis (EDA) was performed on a dataset of 3000 rice leaf images using Roboflow. The dataset includes 12,700 annotations divided into four classes: Brown Spot, Healthy Leaf, Bacterial Blight, and Rice Blast. Most images are square, with a median size of 1600 x 1600 pixels, and are classified as high-resolution 'jumbo' images. Brown Spot has the highest number of annotations (7900), while the other classes have fewer annotations: Healthy Leaf (750), Bacterial Blight (2540), and Rice Blast (1420). The annotation heatmap shows an even distribution across the images, with a slight concentration in the center, and most images have one or two annotated objects. This balanced and diverse dataset ensures a strong foundation for training YOLO models to detect and classify rice leaf diseases effectively.

In this study, a thorough Exploratory Data Analysis (EDA) was conducted using Roboflow to gain insights into the rice leaf disease dataset. This finalized dataset consists of 3000 images, with a total of 12,700 annotations categorized into four classes: Brown Spot, Healthy Leaf, Bacterial Blight, and Rice Blast. The images have an average size of 2.56 megapixels, with a median dimension of 1600 x 1600 pixels, suggesting that most images are square.

Classes

COLOR	CLASS NAME
	Brown Spot
	Healthy Leaf
	Rice Blast
	bacterial blight

Figure 4.1 Dataset classes overview

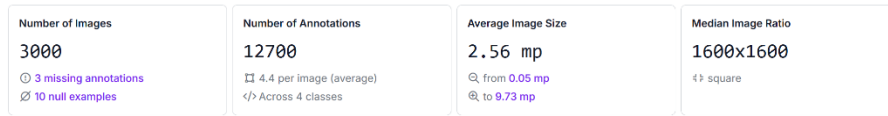


Figure 4.2 Dataset images overview

The class distribution analysis shows that Brown Spot is the most prevalent class with 7900 annotations. In contrast, Healthy Leaf, Bacterial Blight, and Rice Blast have 750, 2540, and 1420 annotations respectively. The relatively low representation of Bacterial Blight and Rice Blast indicates a need for data augmentation to balance the dataset and enhance model accuracy. The size distribution reveals that most images are classified as 'jumbo' (larger than 1024x1024 pixels), ensuring high-resolution data for model training.

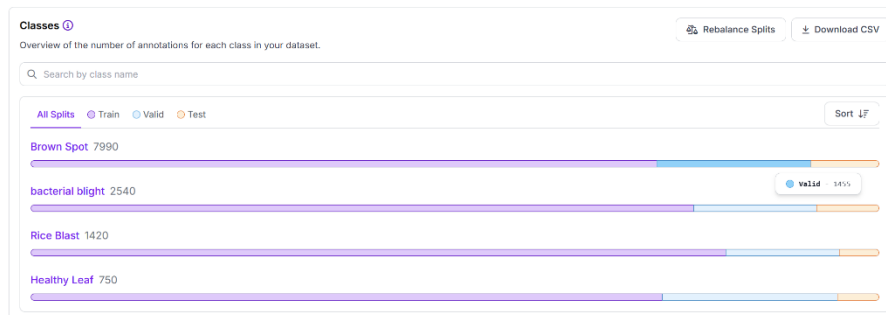


Figure 4.3 Dataset class balance

This dataset dimension image insights image provides an overview of the sizes and aspect ratios of 3,000 images in a dataset. The scatter plot visualizes image dimensions (width vs. height), with most images aligning diagonally, indicating a predominantly square aspect ratio. The dataset is categorized by size: 84.1% of the images are jumbo, 14.5% are small, 1% are medium, and 0.4% are large. In terms of aspect ratio, 97.2% of the images are square, while only 1.1% are tall, 1.7% are wide, and a negligible 0.0% are very wide. This analysis highlights that the dataset is largely uniform in size and aspect ratio, dominated by square and jumbo images, which is useful for understanding dataset consistency and determining preprocessing needs for tasks like machine learning.

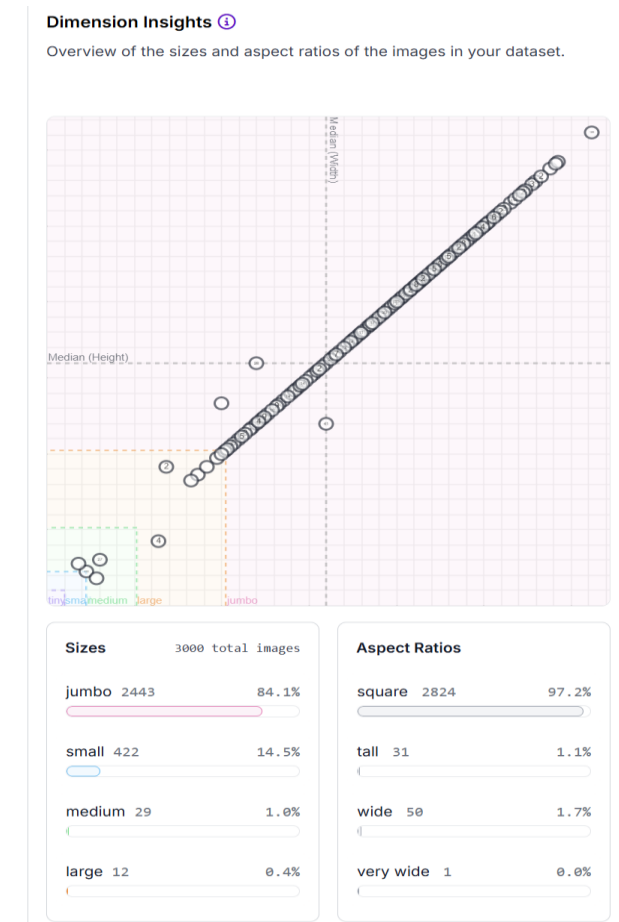


Figure 4.4 Dataset dimension insights

The annotation heatmap reveals that the annotations are evenly spread across the images, with a slight concentration in the center. This even distribution is advantageous for model training as it ensures exposure to varied positions and contexts of the diseases. The histogram of object count per image shows that most images contain one or two annotated objects, with a few images having multiple annotations. This distribution implies that the dataset includes both isolated and clustered instances of rice leaf diseases, providing a rich training environment for the YOLO models.

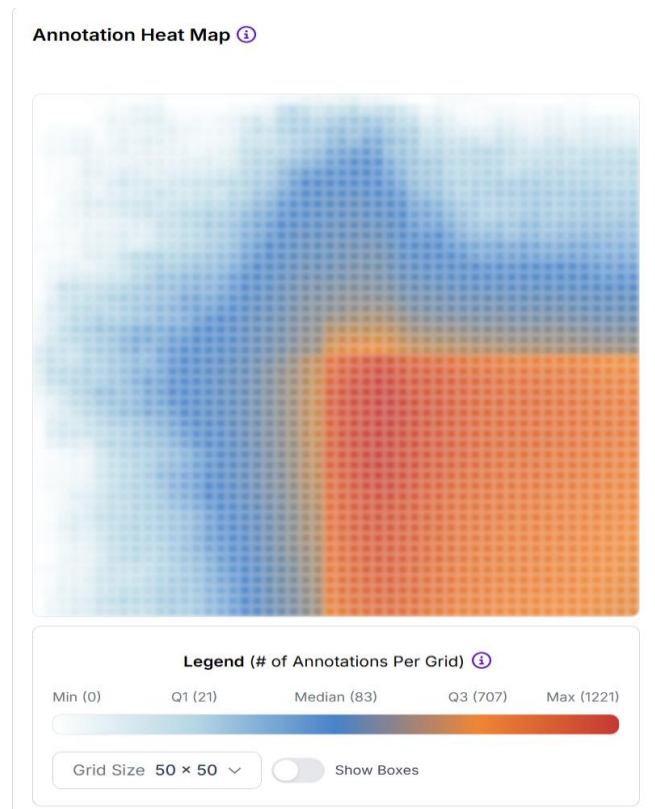


Figure 4.5 Annotation Heatmap

Histogram of Object Count by Image ⓘ

Overview of how many classes are annotated in each image in your dataset.

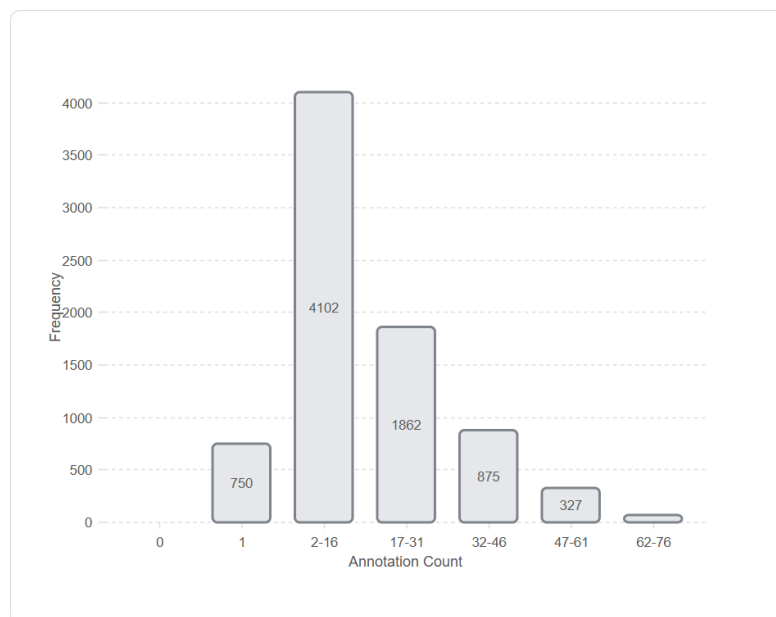


Figure 4.6 Histogram of object count by image

CHAPTER 5

MODEL DEVELOPMENT

5.1 Introduction

This chapter presents how the models were developed for the project. It provides the steps that were taken to design and implement three models for detecting rice leaf diseases, utilizing YOLOv8, YOLOv10, and YOLOv11. Each phase of the development process is explained to provide a detailed understanding of how these models were developed and optimized for disease detection from the images. The development process focuses on data preparation, model training, and evaluation to achieve the best performance for detecting rice leaf diseases.

5.2 Development of the Models:

5.2.1 Training Data:

The models were trained on a curated dataset consisting of 3,000 rice leaf images, annotated into four distinct classes: Healthy Leaf, Bacterial Blight, Brown Spot, and Rice Blast. The images were sourced from open data repositories from Kaggle, and were annotated using the Roboflow tool with the polygon labeling technique. This approach ensured high precision in identifying disease-affected regions. To enhance the dataset's diversity, data augmentation techniques such as random flipping, rotation, scaling, and color adjustments were applied, simulating real-world conditions like variations in lighting, orientation, and scale. The resulting dataset was split into three subsets:

- Training set: 75% of the dataset, used to train the model
- Validation set: 15% was used to monitor performance during training
- Test set: 10% was reserved for evaluating final model performance

5.2.2 Training Process:

The models were trained on Google Colab, leveraging GPU acceleration to significantly enhance computational efficiency and reduce the time required for processing large datasets. This approach allowed for faster training iterations, enabling the models to process and learn from the data more effectively. The training process employed a learning rate of 0.01 to control the step size for parameter updates, ensuring a steady and optimal convergence toward minimizing errors. A total of 100 epochs were executed, with each epoch involving a complete pass through the entire training dataset. The batch size was set to 16, allowing the model to process multiple images simultaneously, which balanced computational efficiency and memory usage.

The optimization of model parameters was carried out using the Stochastic Gradient Descent (SGD) algorithm with momentum. This optimizer not only accelerated convergence but also reduced the risk of getting trapped in local minima by incorporating momentum into the gradient updates. Three key loss functions were utilized during training: classification loss, which measured the model's accuracy in assigning the correct class to detected objects; bounding box regression loss, which assessed the precision of the predicted object boundaries; and objectness loss, which evaluated the confidence scores for the presence of objects within the detected regions. The training was implemented using the PyTorch framework, a widely used deep learning library, and the Ultralytics YOLOv8 library, which provided an optimized implementation of the YOLO algorithm. The annotated dataset, prepared with precise polygon-based labeling, was input into the model. This dataset allowed the model to iteratively refine its parameters through backpropagation, aiming to minimize the defined loss functions and improve detection accuracy.

To ensure the model's generalization capability and prevent overfitting, validation metrics were monitored at the end of each epoch. The validation set, comprising unseen data, provided an unbiased evaluation of the model's performance, allowing for adjustments to the training process if necessary. This systematic and iterative approach ensured that the models were trained effectively and optimized for high accuracy in detecting and classifying rice leaf diseases.

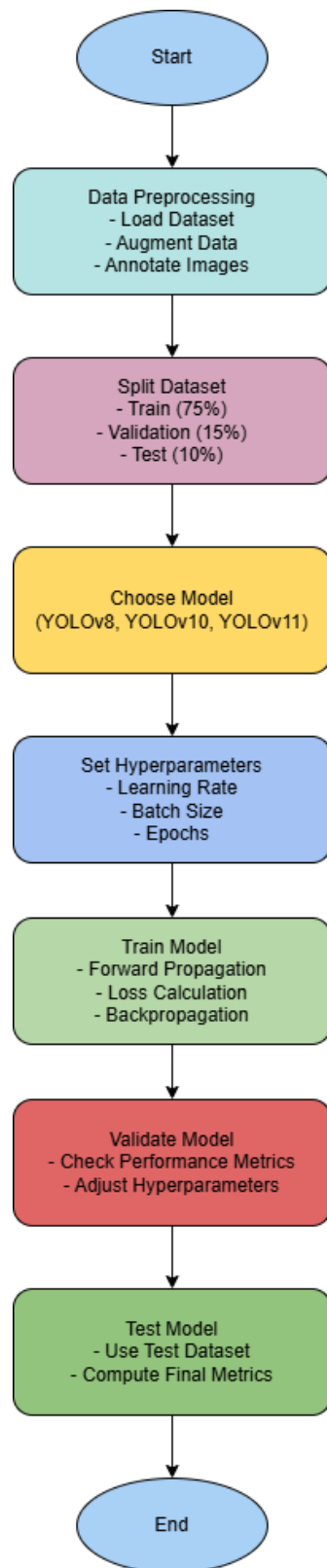


Figure 5.1 Flowchart of the models training process

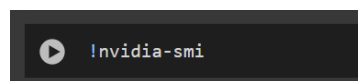
5.2.3 Evaluation Metrics:

The models were evaluated using several performance metrics to assess their effectiveness in detecting and classifying rice leaf diseases. Precision was used to measure the accuracy of predictions, indicating the proportion of correctly identified disease classes among all predicted instances. Recall assessed the model's ability to identify actual positive instances, ensuring comprehensive detection across all classes. The F1-score provided a balanced evaluation by combining precision and recall into a single metric. Additionally, the mean average precision (mAP) was utilized to evaluate the models' overall ability to accurately identify and classify objects, reflecting their reliability and robustness in real-world applications. These metrics collectively demonstrate the effectiveness of the models in addressing the challenges of rice leaf disease detection.

5.3 YOLOv8 Model

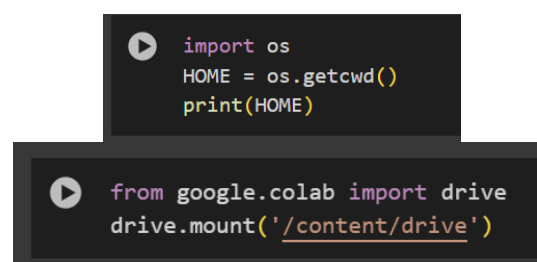
Preparation of the Environment:

At first, it was checked that if the system supports GPU acceleration for faster training. After that, the working directory was set and mounted from the Google Drive.



```
!nvidia-smi
```

Figure 5.2 Checking GPU availability



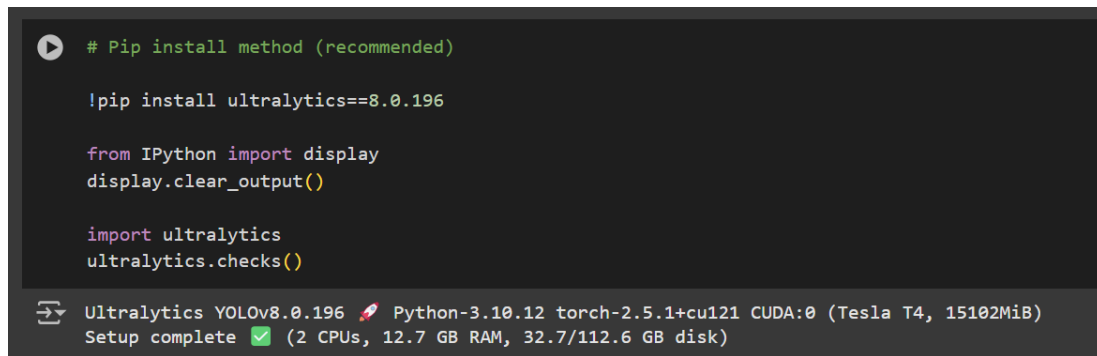
```
import os
HOME = os.getcwd()
print(HOME)

from google.colab import drive
drive.mount('/content/drive')
```

Figure 5.3 Mounting Google Drive

Installing YOLOv8 and importing libraries:

The following command was used to install the YOLOv8 ultralytics library, which is the official library for YOLOv8. The required libraries were imported for the development of the model.



```
# Pip install method (recommended)

!pip install ultralytics==8.0.196

from IPython import display
display.clear_output()

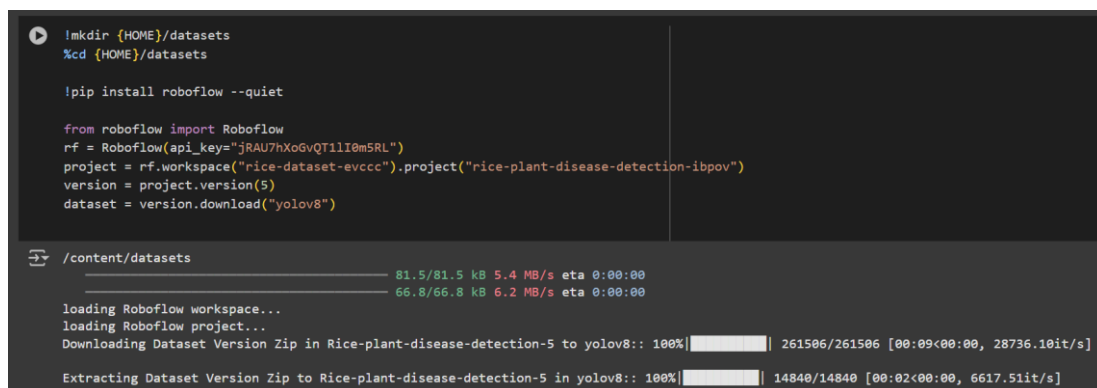
import ultralytics
ultralytics.checks()
```

Ultralytics YOLOv8.0.196 Python-3.10.12 torch-2.5.1+cu121 CUDA:0 (Tesla T4, 15102MiB)
Setup complete (2 CPUs, 12.7 GB RAM, 32.7/112.6 GB disk)

Figure 5.4 Installing and verifying YOLOv8 ultralytics library

Load and Train the YOLOv8 Model:

The YOLOv8 model was loaded and then trained from the organized dataset. The dataset was directly taken from Roboflow annotated dataset.



```
mkdir {HOME}/datasets
%cd {HOME}/datasets

!pip install roboflow --quiet

from roboflow import Roboflow
rf = Roboflow(api_key="jRAU7hXoGvQT1lI0m5RL")
project = rf.workspace("rice-dataset-evccc").project("rice-plant-disease-detection-ibpov")
version = project.version(5)
dataset = version.download("yolov8")
```

/content/datasets
81.5/81.5 kB 5.4 MB/s eta 0:00:00
66.8/66.8 kB 6.2 MB/s eta 0:00:00
loading Roboflow workspace...
loading Roboflow project...
Downloading Dataset Version Zip in Rice-plant-disease-detection-5 to yolov8:: 100% | 261506/261506 [00:09<00:00, 28736.10it/s]
Extracting Dataset Version Zip to Rice-plant-disease-detection-5 in yolov8:: 100% | 14840/14840 [00:02<00:00, 6617.51it/s]

Figure 5.5 YOLOv8 installation and setup check

Custom training and evaluating the model:

Then, the model was trained by fine-tuning the hyperparameters. After training, the model's performance was evaluated on the validation dataset:

```
Custom Training

[ ] # Hyperparameter Tuning CLI
%cd {HOME}

!yolo task=detect mode=train model=yolov8m.pt data={dataset.location}/data.yaml epochs=100 imgsz=640 optimizer="AdamW" plots=True

metrics = model.val()
print(metrics)
```

Figure 5.6 Custom training and evaluating the model

Finalized trained model's output:

The first line lists the contents of the YOLOv8 training results folder, showing files like model weights (best.pt) and validation results. The second line changes to the working directory. It displays a validation prediction image (val_batch0_pred.jpg), showcasing the model's bounding boxes, class labels, and confidence scores to visually evaluate detection performance.

```
!ls {HOME}/runs/detect/train/

args.yaml          R_curve.png       val_batch0_labels.jpg
confusion_matrix_normalized.png  results.csv       val_batch0_pred.jpg
confusion_matrix.png      results.png       val_batch1_labels.jpg
events.out.tfevents.1734492592.605c28bc81d9.1053.0  train_batch0.jpg  val_batch1_pred.jpg
F1_curve.png          train_batch1.jpg  val_batch2_labels.jpg
labels_correlogram.jpg  train_batch2.jpg  val_batch2_pred.jpg
labels.jpg            train_batch38070.jpg  weights
P_curve.png           train_batch38071.jpg
PR_curve.png          train_batch38072.jpg

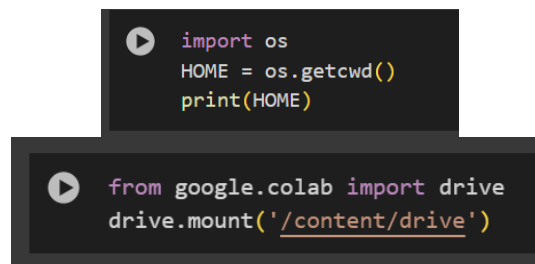
%cd {HOME}
Image(filename=f'{HOME}/runs/detect/train/val_batch0_pred.jpg', width=600)
```

Figure 5.7 Creation of training results and visualizations

5.4 YOLOv10 Model

Preparation of the Environment:

At first, it was checked that if the system supports GPU acceleration for faster training. After that the working directory was set, mounted from the google drive.



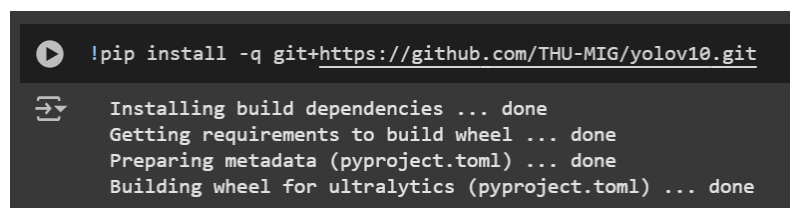
```
import os
HOME = os.getcwd()
print(HOME)
```

```
from google.colab import drive
drive.mount('/content/drive')
```

Figure 5.8 Mounting Google Drive

Installing YOLOv10 and importing libraries:

The following command was used to install the YOLOv10 ultralytics library, which is the official library for YOLOv10. The required libraries were imported for the development of the model.



```
!pip install -q git+https://github.com/THU-MIG/yolov10.git
```

```
Installing build dependencies ... done
Getting requirements to build wheel ... done
Preparing metadata (pyproject.toml) ... done
Building wheel for ultralytics (pyproject.toml) ... done
```

Figure 5.9 Installation of YOLOv10 ultralytics library

Load and Train the YOLOv10 Model:

The YOLOv10 model was loaded and then trained from the organized dataset. The dataset was directly taken from Roboflow annotated dataset.

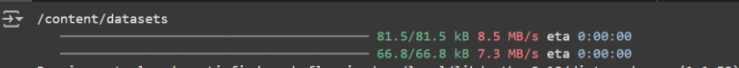
```
!mkdir {HOME}/datasets
%cd {HOME}/datasets

!pip install -q roboflow

from google.colab import userdata
from roboflow import Roboflow

!pip install roboflow

from roboflow import Roboflow
rf = Roboflow(api_key="jRAU7hXoGvQT1lI0m5RL")
project = rf.workspace("rice-dataset-evccc").project("rice-plant-disease-detection-ibpov")
version = project.version(5)
dataset = version.download("yolov10")
```



The Colab file browser shows the directory /content/datasets. It contains two files: a file named 81.5/81.5 kB 8.5 MB/s eta 0:00:00 and a file named 66.8/66.8 kB 7.3 MB/s eta 0:00:00.

Figure 5.10 YOLOv10 model setup and Roboflow dataset integration

Custom training and evaluating the model:

Then, the model was trained by fine-tuning the hyperparameters. After training, the model's performance was evaluated on the validation dataset:

```
%cd {HOME}

!yolo task=detect mode=train epochs=100 batch=32 plots=True \
model={HOME}/weights/yolov10m.pt \
data={dataset.location}/data.yaml

metrics = model.val()
print(metrics)
```

Figure 5.11 Custom training and evaluating the model

Finalised trained model's output:

The first line lists the contents of the YOLOv10 training results folder, showing files like model weights (best.pt) and validation results. The second line changes to the working directory. It displays a validation prediction image (val_batch0_pred.jpg), showcasing the model's bounding boxes, class labels, and confidence scores to visually evaluate detection performance.

```
[ ] !pip install supervisely
    from ultralytics import YOLOv10
    import supervisely as sv

    model = YOLOv10(f'{HOME}/runs/detect/train/weights/best.pt')

    dataset = sv.DetectionDataset.from_yolo(
        images_directory_path=f'{dataset.location}/valid/images",
        annotations_directory_path=f'{dataset.location}/valid/labels"
        data_yaml_path=f'{dataset.location}/data.yaml"
    )

    bounding_box_annotator = sv.BoundingBoxAnnotator()
    label_annotator = sv.LabelAnnotator()
```

```
!ls {HOME}/runs/detect/train/
```

args.yaml	R_curve.png	val_batch0_labels.jpg
confusion_matrix_normalized.png	results.csv	val_batch0_pred.jpg
confusion_matrix.png	results.png	val_batch1_labels.jpg
events.out.tfevents.1734492592.605c28bc81d9.1053.0	train_batch0.jpg	val_batch1_pred.jpg
F1_curve.png	train_batch1.jpg	val_batch2_labels.jpg
labels_correlogram.jpg	train_batch2.jpg	val_batch2_pred.jpg
labels.jpg	train_batch38070.jpg	weights
P_curve.png	train_batch38071.jpg	
PR_curve.png	train_batch38072.jpg	

Figure 5.12 Creation of training results and visualizations

5.5 YOLOv11 Model

Preparation of the Environment:

The working directory was set and mounted from the Google Drive.

```
import os
HOME = os.getcwd()
print(HOME)
```

```
from google.colab import drive
drive.mount('/content/drive')
```

Figure 5.13 Mounting Google Drive

Installing YOLOv11 and importing libraries:

The following command was used to install the YOLOv11 ultralytics library, which is the official library for YOLOv11. The required libraries were imported for the development of the model.

```
[ ] %pip install ultralytics
import ultralytics
ultralytics.checks()

Ultralytics 8.3.52 Python-3.10.12 torch-2.5.1+cu121 CUDA:0 (Tesla T4, 15102MiB)
Setup complete (2 CPUs, 12.7 GB RAM, 32.7/112.6 GB disk)
```

Figure 5.14 Installation of YOLOv11 ultralytics library

Tracker for the YOLOv11 Training Model:

The following allows the user to select a logging tool for tracking the YOLOv11 training process. Comet provides an advanced cloud-based interface, while TensorBoard is a local or cloud-based visualization tool. These tools help monitor training metrics and improve the debugging and optimization of the model.

```
##title Select YOLO11 logger {run: 'auto'}
logger = 'TensorBoard' #@param ['Comet', 'TensorBoard']

if logger == 'Comet':
    %pip install -q comet_ml
    import comet_ml; comet_ml.init()
elif logger == 'TensorBoard':
    %load_ext tensorboard
    %tensorboard --logdir .

TensorBoard
```

Figure 5.15 YOLOv11 training tracker setup with TensorBoard.

Load and Train the YOLOv8 Model:

The YOLOv11 model was loaded and then trained from the organized dataset. The dataset was directly taken from Roboflow annotated dataset.

```

!pip install roboflow

from roboflow import Roboflow
rf = Roboflow(api_key="jRAU7hXoGvQT1lI0m5RL")
project = rf.workspace("rice-dataset-evccc").project("rice-plant-disease-detection-ibpov")
version = project.version(5)
dataset = version.download("yolov11")

```

Figure 5.16 YOLOv11 model setup and Roboflow dataset integration

Custom training and evaluating the model:

Then, the model was trained by fine-tuning the hyperparameters. After training, the model's performance was evaluated on the validation dataset:

```

# Train YOLO11m Custom Model
!yolo train model=yolov11m.pt data=/content/Rice-plant-disease-detection-5/data.yaml epochs=100 imgsz=640

metrics = model.val()
print(metrics)

```

Figure 5.17 Custom training and evaluating the model

Finalized trained model's output:

The first line lists the contents of the YOLOv8 training results folder, showing files like model weights (best.pt) and validation results. The second line changes to the working directory. It displays a validation prediction image (val_batch0_pred.jpg), showcasing the model's bounding boxes, class labels, and confidence scores to visually evaluate detection performance.

```

!ls /content/runs/detect/train

args.yaml          R_curve.png       val_batch0_labels.jpg
confusion_matrix_normalized.png  results.csv       val_batch0_pred.jpg
confusion_matrix.png  results.png       val_batch1_labels.jpg
events.out.tfevents.1734779302.5605f12431bb.3041.0  train_batch0.jpg  val_batch1_pred.jpg
F1_curve.png        train_batch1.jpg  val_batch2_labels.jpg
labels_correlogram.jpg  train_batch2.jpg  val_batch2_pred.jpg
labels.jpg           train_batch38070.jpg  weights
P_curve.png          train_batch38071.jpg
PR_curve.png         train_batch38072.jpg

```

Figure 5.18 Creation of training results and visualizations

5.6 Summary

Chapter 5 highlights the detailed development process of the models for detecting rice leaf diseases using three versions of the YOLO algorithm: YOLOv8, YOLOv10, and YOLOv11. It outlines the coding sections and systematic process of developing and evaluating YOLO-based models for rice leaf disease detection, focusing their potential for high accuracy and efficiency.

CHAPTER 6

RESULTS AND DASHBOARD

6.1 Introduction

This chapter describes the various phases of the proposed solution, as explained in Chapter 3. It further elaborates on the results, step by step, and output of the trained models – YOLOv08, YOLOv10 and YOLOv11, along with the outcomes of the finalized dashboard so that a clear understanding of the whole process can be gained.

6.2 Results based on the YOLOv8 Model

In this section, we present and analyze the results obtained from the YOLOv8 model to detect rice leaf diseases. The performance of the YOLOv8 model is evaluated on the custom dataset, which includes training, validation, and testing datasets. The key metrics used for this evaluation include precision, recall, loss, and the confusion matrix.

6.2.1 Precision

The outcomes of training the custom rice leaf disease dataset using the YOLOv8 model are impressive with a precision of more than 80%. The dataset was produced by gathering images of diverse rice leaf diseases and carefully annotating them with polygon-based labelling tool in Roboflow, a technique that ensures precise representation of complex disease patterns. The YOLOv8 model, recognised for its refined architecture and effective optimization techniques, was subsequently trained on these labelled images. Data augmentation methods were utilized during the 100-epoch training procedure to improve the model's generalization capabilities. The model achieved a precision rate of more than 80%, illustrating its effectiveness in

identifying different rice leaf diseases from healthy leaves. Such a high level of precision is critical for early detection and precise treatment of leaf diseases in agriculture, thereby enhancing better crop management and higher yields.

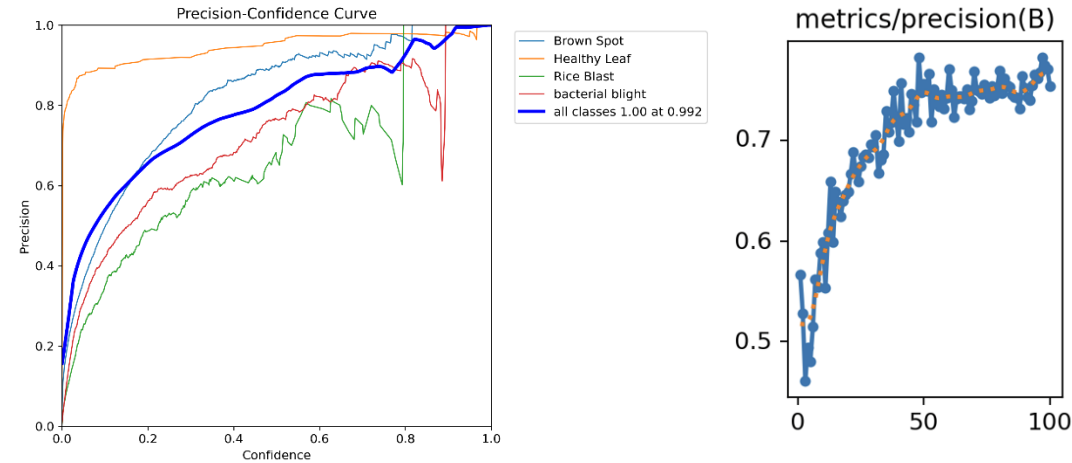


Figure 6.1 Precision-Confidence Curve based on the YOLOv8 model

6.2.2 Precision Recall

This Precision-Recall curve shows how well the YOLOv8 model can recognize different conditions in rice leaves. Each line on the graph represents a specific class—Brown Spot, Healthy Leaf, Rice Blast, and Bacterial Blight—while the thicker blue line combines all classes. The overall mAP@0.5 is 0.880, which means the model is quite accurate. Healthy Leaf (orange line) achieves the best performance at 0.944, indicating the model easily tells healthy leaves apart from diseased ones. Brown Spot (0.872), Bacterial Blight (0.863), and Rice Blast (0.842) also show strong scores. This suggests the model can reliably detect and classify different leaf conditions, which is important for early disease detection and better crop management.

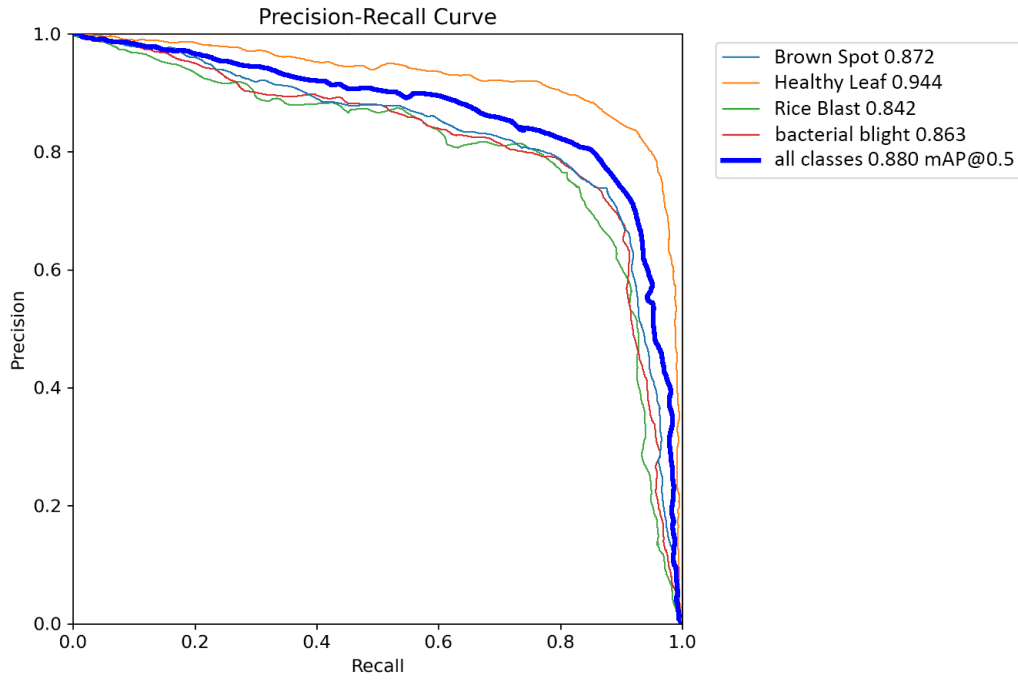


Figure 6.2 Precision-Recall Curve based on the YOLOv8 model

6.2.3 F1-Confidence Curve

The F1-Confidence Curve illustrates the performance of a classification model for detecting various rice leaf conditions, with the x-axis representing confidence thresholds and the y-axis indicating F1 scores. The model evaluates four classes: Brown Spot, Healthy Leaf, Rice Blast, and Bacterial Blight, along with an overall performance curve for all classes. The optimal confidence threshold for all classes is identified at 0.275, achieving an F1 score of 0.80, as represented by the bold blue line. Among individual classes, Healthy Leaf shows the highest F1 scores across a wide range of confidence levels, indicating strong classification performance, followed by Brown Spot. In contrast, Bacterial Blight and Rice Blast have relatively lower F1 scores, suggesting room for improvement in detecting these conditions. Overall, the graph highlights the model's strengths in classifying certain conditions while providing insights for optimizing its performance across all categories.

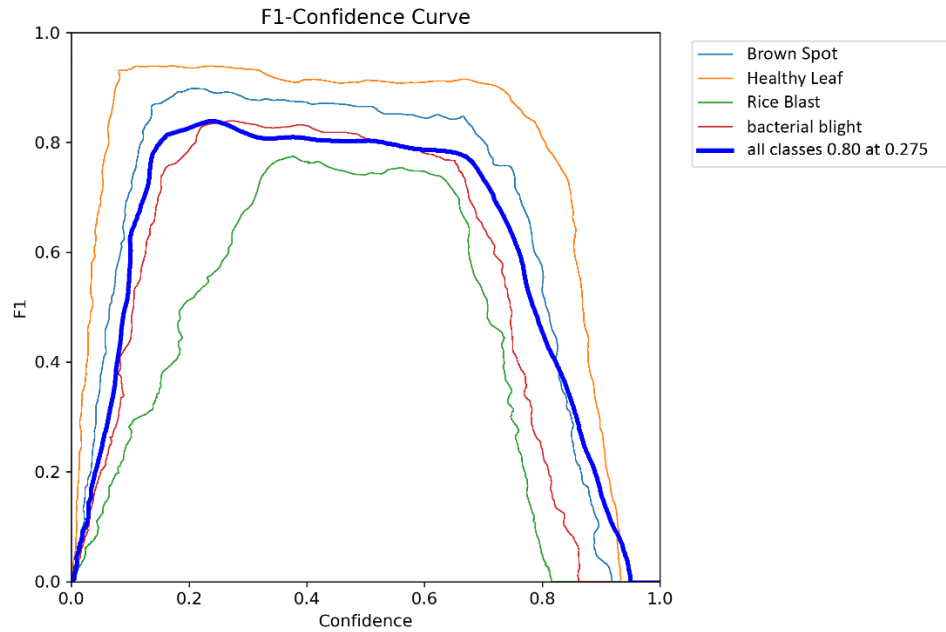


Figure 6.3 F1-Confidence Curve based on the YOLOv8 model

6.2.4 Confusion Metrics

This confusion matrix displays how well the YOLOv8 model distinguishes between Brown Spot, Healthy Leaf, Rice Blast, Bacterial Blight, and background. The diagonal cells in dark blue indicate correct predictions, while the off-diagonal cells represent misclassifications. Brown Spot is correctly identified 83% of the time, with 17% confused as background. Healthy Leaf has the highest accuracy at 92%, but 8% still gets mislabeled as background. Rice Blast has an 80% correct rate, with the remaining 20% mixed up as background. Bacterial Blight is detected 81% of the time, while 19% is wrongly classified as background. Although the model generally performs well, these results suggest that further tuning or more training data could help reduce the confusion between diseased leaves and background, thereby increasing overall accuracy.

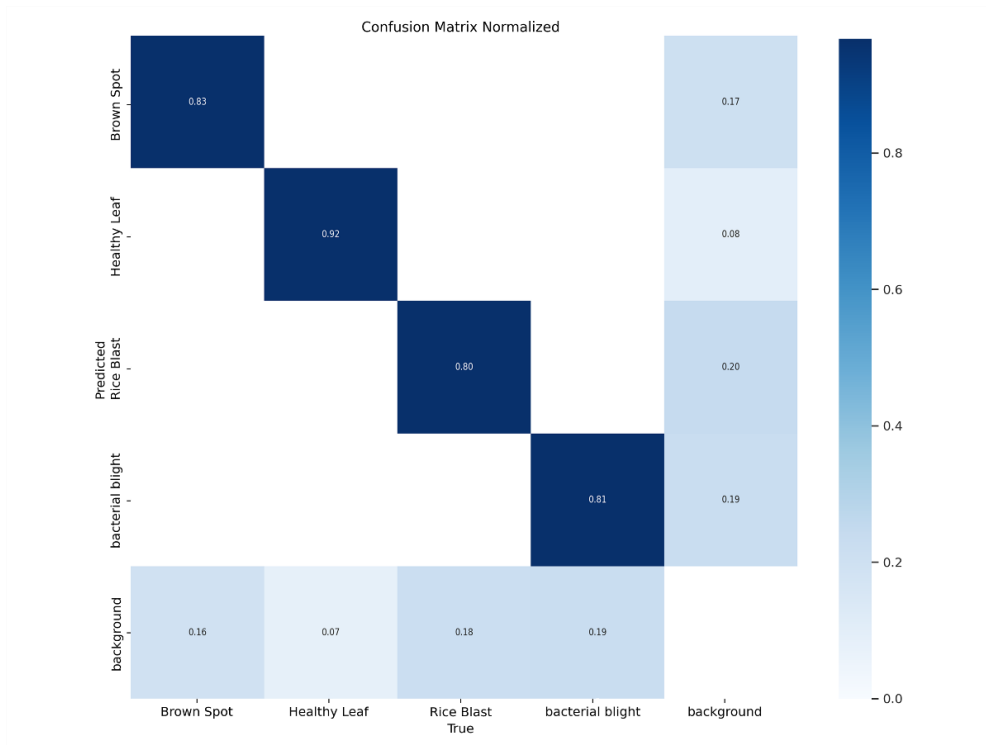


Figure 6.4 Confusion matrix based on the YOLOv8 model

6.2.5 Loss Metrics

These three loss metrics graphs show how the YOLOv8 model's loss values get smaller over 100 training epochs. The first chart (train/box_loss) tracks the bounding-box error, the second chart (train/cls_loss) measures how well the model classifies each object, and the third chart (train/dfl_loss) is another measure related to bounding-box quality. At the start, all losses are high, but they quickly drop and then keep going down more slowly. By the end of the 100 epochs, the losses become quite low, which means the model is getting better at both finding and classifying the rice leaf diseases in the dataset.

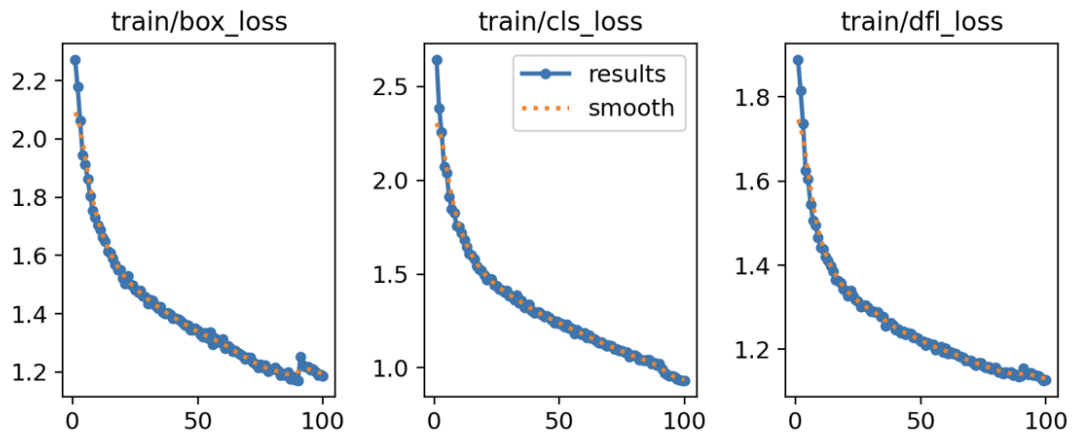


Figure 6.5 Loss metrics graph based on the YOLOv8 model

6.2.6 Results on Testing Images using the YOLOv8 Model

This section discusses the performance of the YOLOv8 model on testing images. The images given in the following were tested using the YOLOv8 model. Figures 6.5 to 6.8 display the detection results, including confidence levels, for all classes of rice leaf diseases.

Detection of Healthy Leaves

The detection results for healthy leaves are presented in Figure 4.5. The YOLOv8 model correctly identified the healthy leaves, highlighting its capability to differentiate between diseased and healthy plant parts.

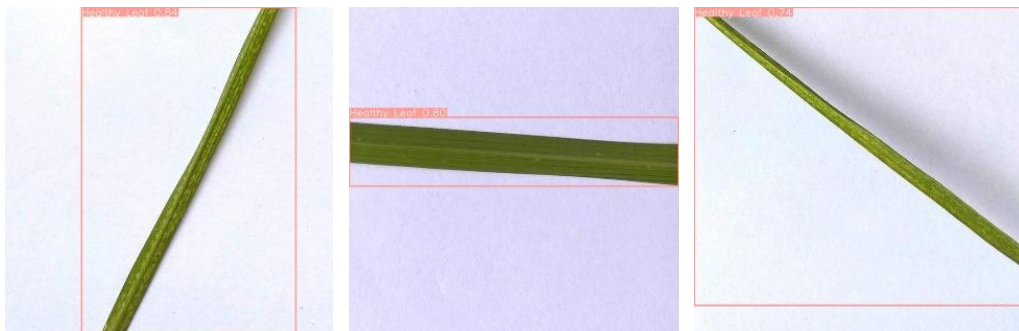


Figure 6.6 Detection for Healthy Leaves

Detection of Bacterial Blight

Figure 6.6 illustrates the detection of bacterial leaf blight. The YOLOv8 model successfully localized and classified the bacterial blight diseases, showcasing its effectiveness in identifying this disease under various conditions.



Figure 6.7 Detection for Bacterial Blights

Detection of Brown Spots

The YOLOv8 model successfully detected brown spot diseases on rice leaves, as shown in Figure 6.7. The model's predictions are marked with bounding boxes and corresponding confidence scores, demonstrating its ability to detect brown spot with high precision.

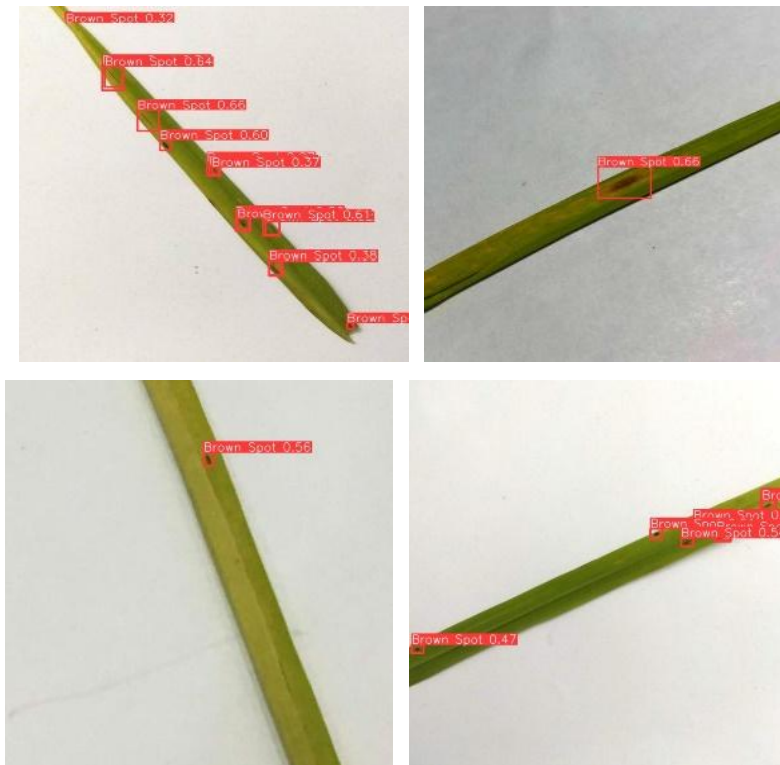


Figure 6.8 Detection for Brown Spots

Detection of Rice Blasts

The detection of blast disease on rice leaves is illustrated in Figure 4.8. The performance of the YOLOv8 model in detecting rice blasts is shown by the bounding boxes and confidence levels, indicating its robustness in identifying this specific disease.

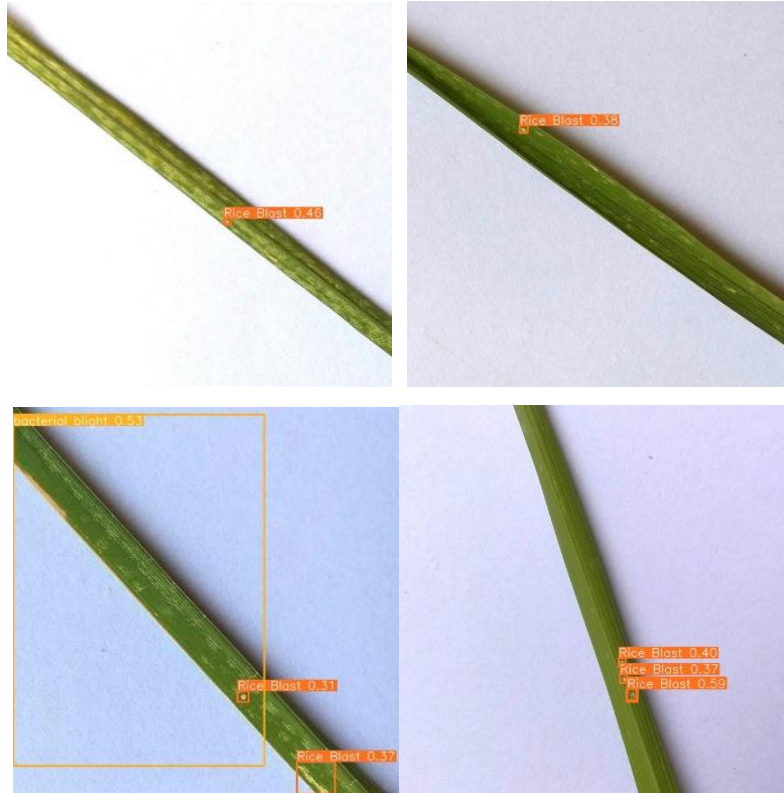


Figure 6.9 Detection for Rice Blasts

6.3 Results based on the YOLOv10 Model

In this section, we present and analyze the results obtained from the YOLOv10 model to detect rice leaf diseases. The performance of the YOLOv10 model is evaluated on the custom dataset, which includes training, validation, and testing datasets. The key metrics used for this evaluation include precision, recall, loss, and the confusion matrix.

6.3.1 Precision

Below are the YOLOv10 precision results for detecting different rice leaf conditions. The graph shows how precision changes as the model's confidence threshold goes up. Each colored line represents a different class—Brown Spot, Healthy

Leaf, Rice Blast, and Bacterial Blight—while the thick blue line sums up all classes. The Healthy Leaf (orange line) keeps a high precision across nearly the entire range of confidence values, while Brown Spot, Bacterial Blight, and Rice Blast also rise steadily before leveling off at higher thresholds. In the second graph, The second graph it shows how the model’s average precision evolves over 100 epochs of training. Starting around 0.5, the mAP climbs to about 90%, showing a clear improvement in the model’s ability to detect diseases accurately over time. By the final epochs, YOLOv10 demonstrates strong performance in distinguishing between healthy leaves and various rice leaf diseases, as indicated by high precision scores.

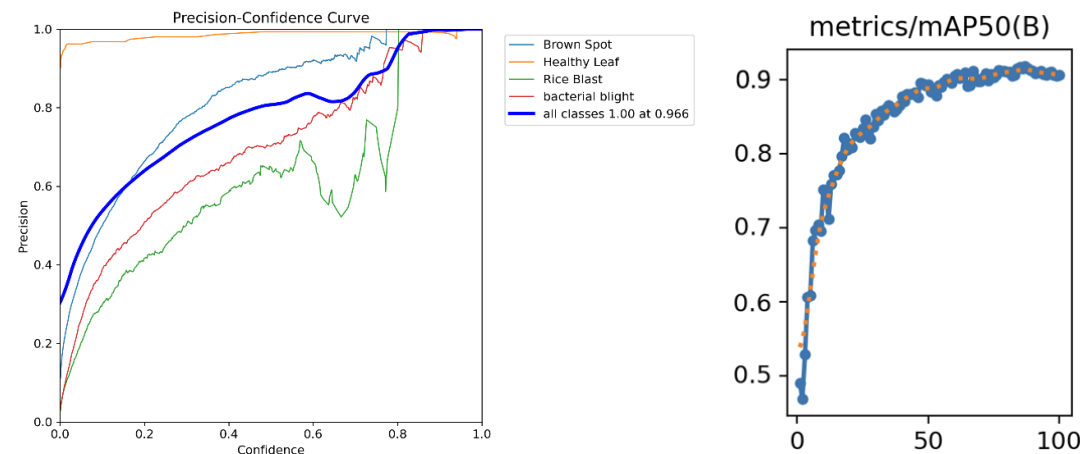


Figure 6.10 Precision-Confidence Curve based on the YOLOv10 model

6.3.2 Precision Recall

This Precision-Recall curve shows how well the YOLOv10 model can distinguish various rice leaf conditions. Each colored line represents a different class—Brown Spot, Healthy Leaf, Rice Blast, and Bacterial Blight—while the thick blue line is the overall performance for all classes. The area under each curve reflects the balance between precision (y-axis) and recall (x-axis): higher and more consistent lines indicate better detection quality. Healthy Leaf (orange line) maintains the highest performance at 0.963, while the other three diseases also produce strong scores above 0.85. Overall, the model achieves an mAP@0.5 of 0.899, suggesting that YOLOv10

is quite effective at correctly identifying diseased and healthy leaves across a range of recall values.

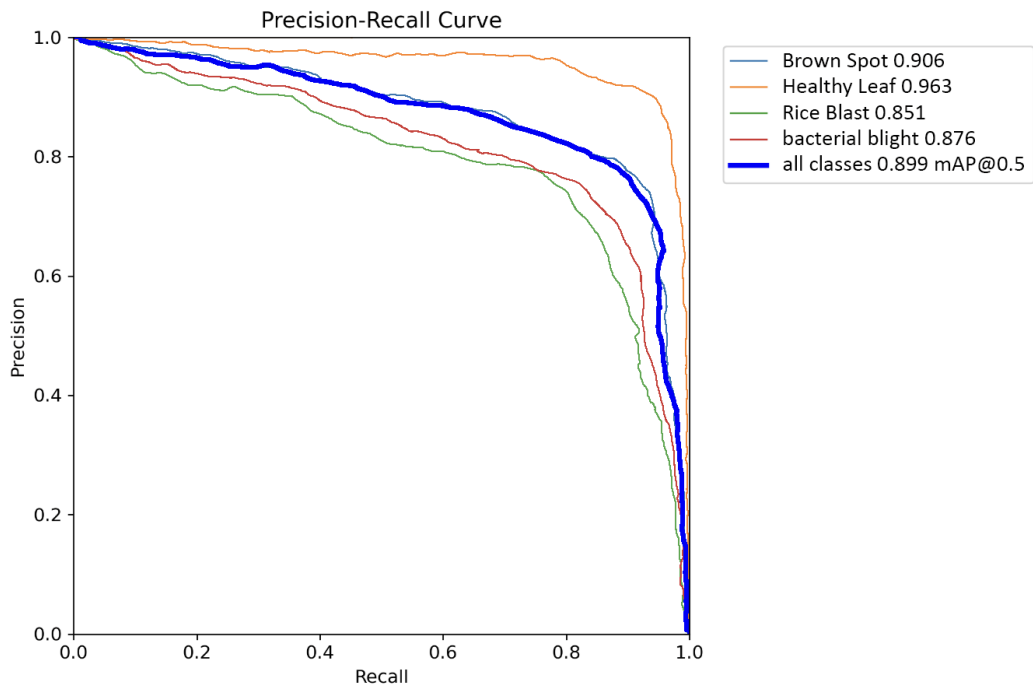


Figure 6.11 Precision-Recall Curve based on the YOLOv10 model

6.3.3 F1-Confidence Curve

The F1-Confidence Curve for the YOLOv10 model illustrates the model's performance in detecting four categories of rice leaf conditions: Brown Spot, Healthy Leaf, Rice Blast, and Bacterial Blight. The x-axis represents the confidence threshold, while the y-axis indicates the F1 score, which balances precision and recall. The optimal performance for all classes occurs at a confidence threshold of 0.275, achieving an overall F1 score of 0.85, as shown by the bold blue line. Among individual classes, Healthy Leaf achieves the highest F1 score across a broad confidence range, indicating strong classification accuracy. Brown Spot also shows robust performance, while Bacterial Blight and Rice Blast exhibit relatively lower F1 scores, suggesting areas for potential improvement. This curve highlights the

YOLOv10 model's ability to generalize well for certain categories while providing insights into optimizing its thresholds for balanced classification performance.

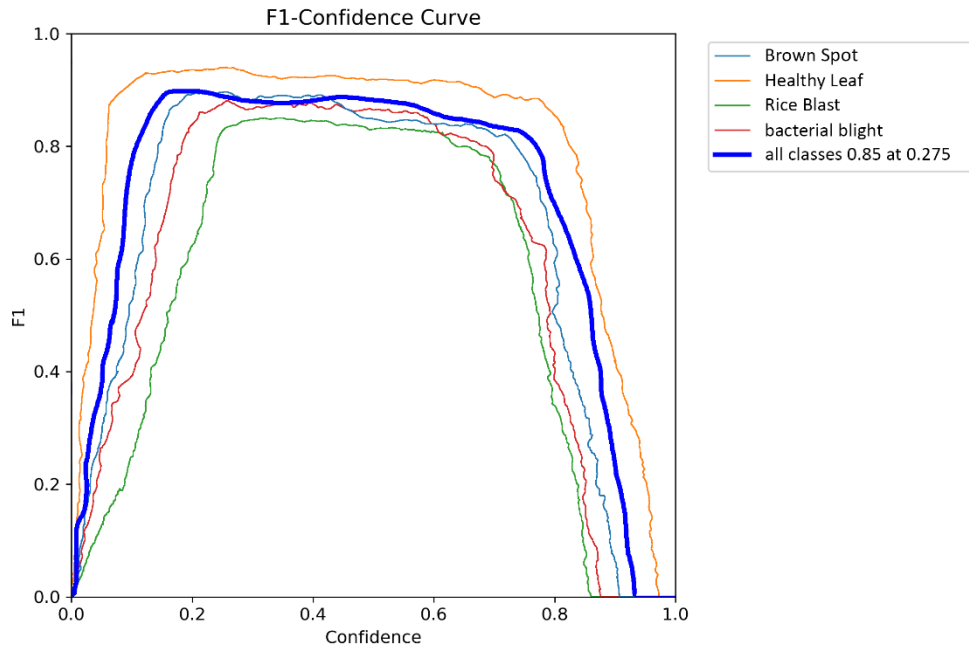


Figure 6.12 F1-Confidence Curve based on the YOLOv10 model

6.3.4 Confusion Metrics

This confusion matrix displays how well the YOLOv10 model distinguishes between Brown Spot, Healthy Leaf, Rice Blast, Bacterial Blight, and background. The diagonal cells in dark blue indicate correct predictions, while the off-diagonal cells represent misclassifications. Brown Spot is correctly identified 87% of the time, with 13% confused as background. Healthy Leaf has the highest accuracy at 95%, but 5% still gets mislabeled as background. Rice Blast has an 82% correct rate, with the remaining 18% mixed up as background. Bacterial Blight is detected 83% of the time, while 17% is wrongly classified as background. Although the model generally performs well, these results suggest that further tuning or additional training data could help reduce the confusion between diseased leaves and background, thereby increasing overall accuracy.

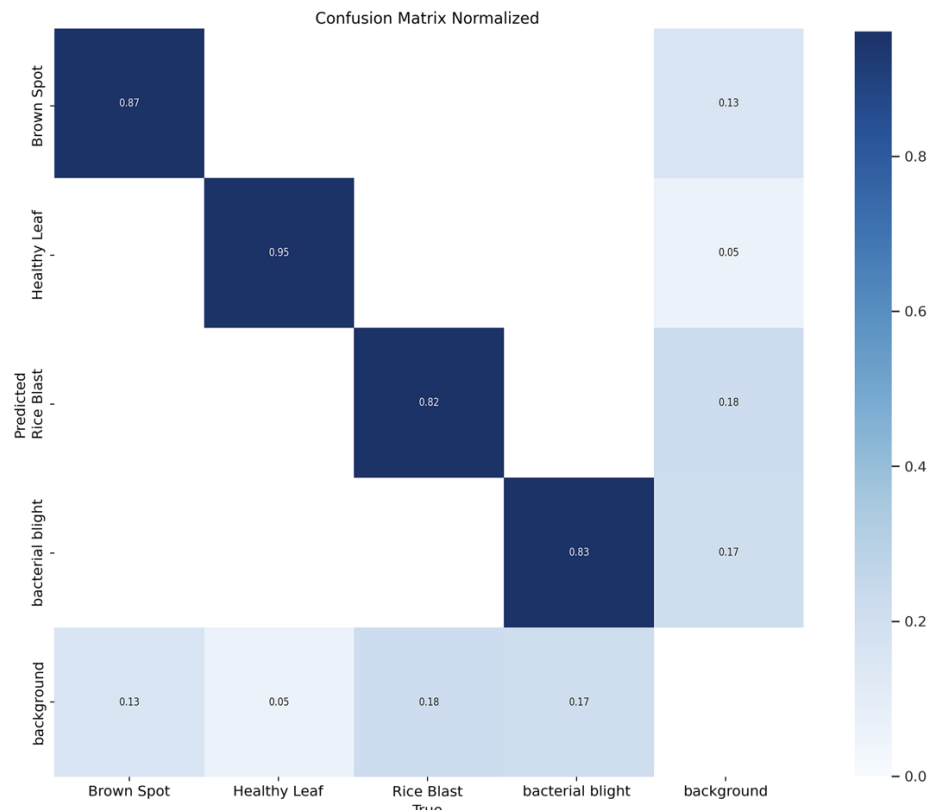


Figure 6.13 Confusion matrix based on the YOLOv10 model

6.3.5 Loss Metrics

The following graphs shows how the YOLOv10 model's losses go down over 100 training epochs. Each chart focuses on a different loss type—box, classification, and distribution focal loss—indicating how well the model is learning to localize and classify rice leaf diseases. Initially, all three losses are quite high, but they quickly drop during the early epochs and then decrease more gradually. By the end of training, each loss has settled at a much lower level, suggesting the model has effectively learned to detect and differentiate between diseased and healthy leaves.

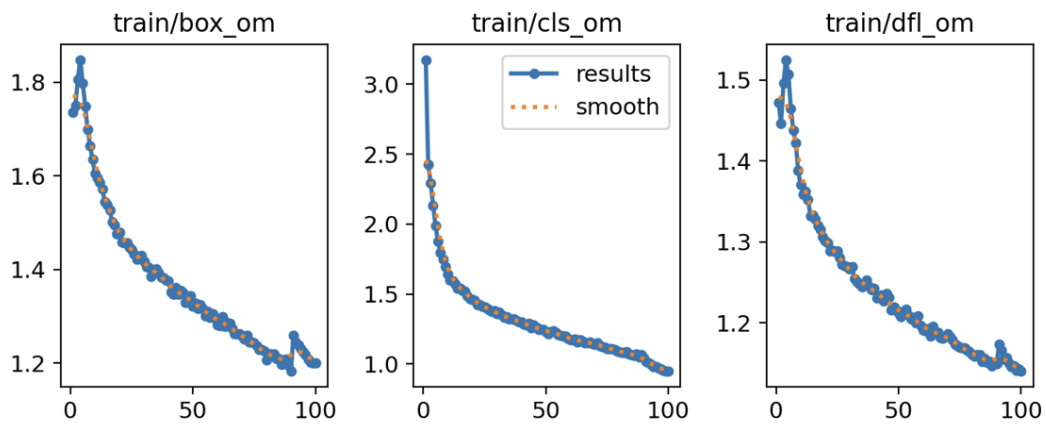


Figure 6.14 Loss metrics graph based on the YOLOv10 model

6.3.6 Results on Testing Images using the YOLOv8 Model

This section discusses the performance of the YOLOv10 model on testing images. The images given in the following were tested using the YOLOv10 model. Figures 6.13 to 6.15 display the detection results, including confidence levels, for all classes of rice leaf diseases.

Detection of Healthy Leaves

The detection results for healthy leaves are presented in Figure 6.13. The YOLOv10 model correctly identified the healthy leaves, highlighting its capability to differentiate between diseased and healthy plant parts.

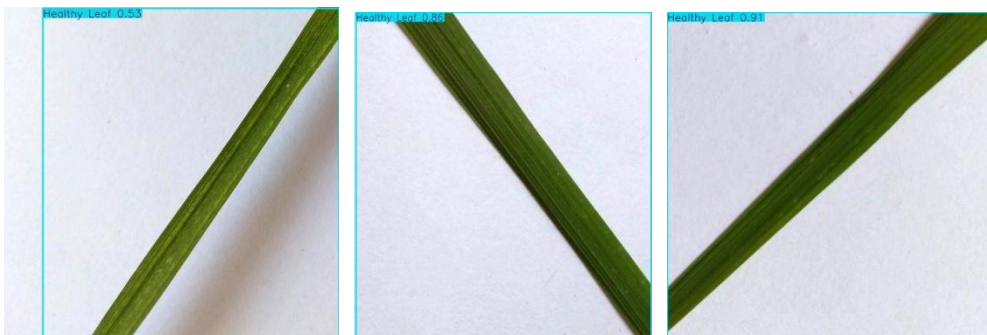


Figure 6.15 Detection for Healthy Leaves

Detection of Bacterial Blight

Figure 6.14 illustrates the detection of bacterial leaf blight. The YOLOv10 model successfully localized and classified the bacterial blight diseases, showcasing its effectiveness in identifying this disease under various conditions.



Figure 6.16 Detection for Bacterial Blights

Detection of Brown Spots

The YOLOv10 model successfully detected brown spot diseases on rice leaves, as shown in Figure 6.15. The model's predictions are marked with bounding boxes and corresponding confidence scores, demonstrating its ability to detect brown spot with high precision.

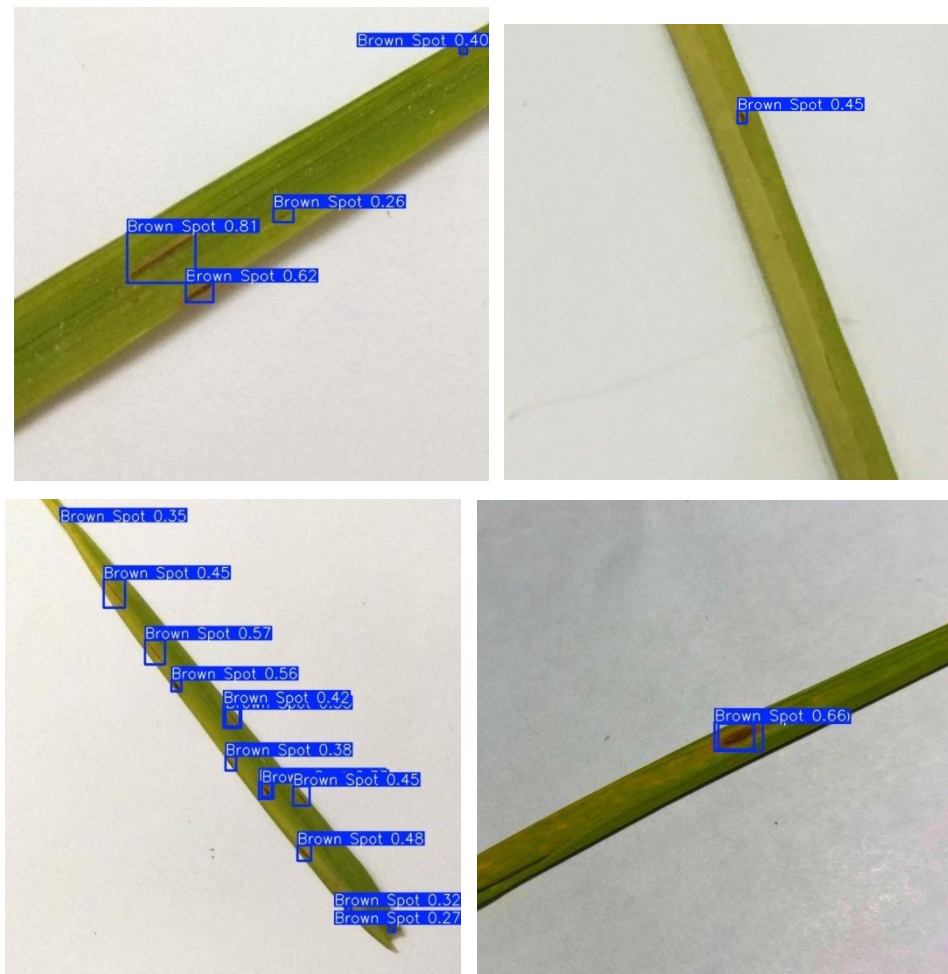


Figure 6.17 Detection for Brown Spots

Detection of Rice Blasts

The detection of blast disease on rice leaves is illustrated in Figure 6.16. The performance of the YOLOv10 model in detecting rice blasts is shown by the bounding boxes and confidence levels, indicating its robustness in identifying this specific disease.

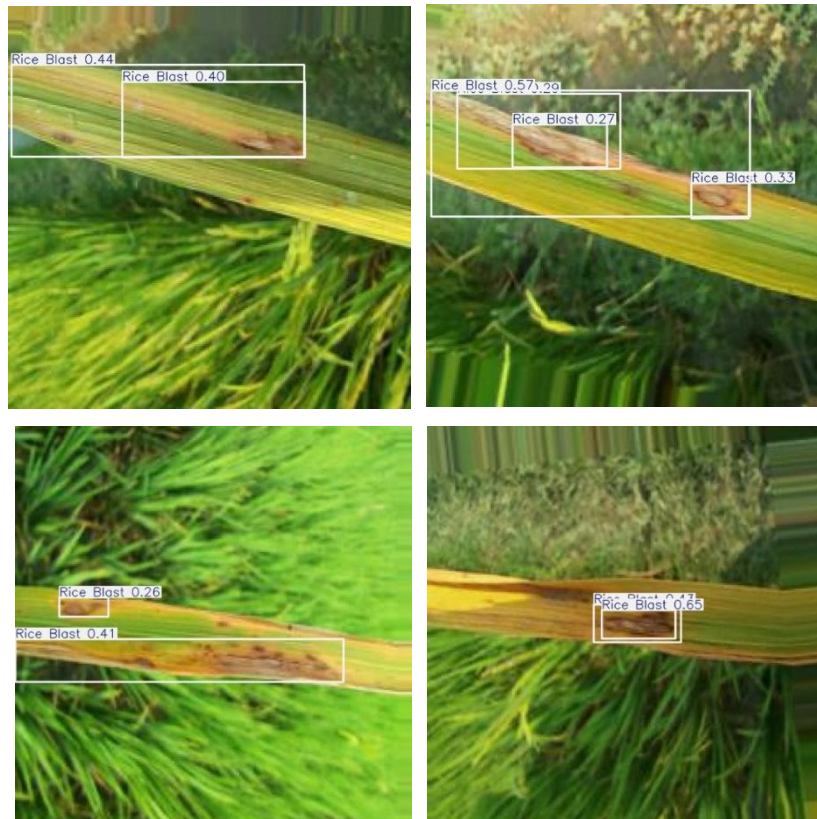


Figure 6.18 Detection for Rice Blasts

6.4 Results based on the YOLOv11 Model

In this section, we present and analyze the results obtained from the YOLOv11 model to detect rice leaf diseases. The performance of the YOLOv11 model is evaluated on the custom dataset, which includes training, validation, and testing datasets. The key metrics used for this evaluation include precision, recall, loss, and the confusion matrix.

6.4.1 Precision

Below are the YOLOv11 precision results for detecting different rice leaf conditions, shown in two charts. The first chart (Precision-Confidence Curve) illustrates how precision rises or falls as the confidence threshold changes. Each

colored line corresponds to a specific disease class—Brown Spot (blue), Healthy Leaf (orange), Rice Blast (green), and Bacterial Blight (red)—while the thick blue line represents the combined performance for all classes. Notably, Healthy Leaf retains very high precision across a broad range of confidence levels, and the model’s overall precision (the blue line) reaches 1.00 at around 0.935 confidence, indicating strong classification accuracy for all classes. The second chart (metrics/precision(B)) shows how the model’s precision progresses over 100 epochs. It starts at a moderate level and climbs steadily, eventually crossing 90% in the later stages. This upward trend suggests that YOLOv11 is effectively learning to differentiate healthy and diseased leaves as training continues. By the end of these 100 epochs, the model demonstrates strong precision results, reflecting its capability to accurately identify various rice leaf conditions.

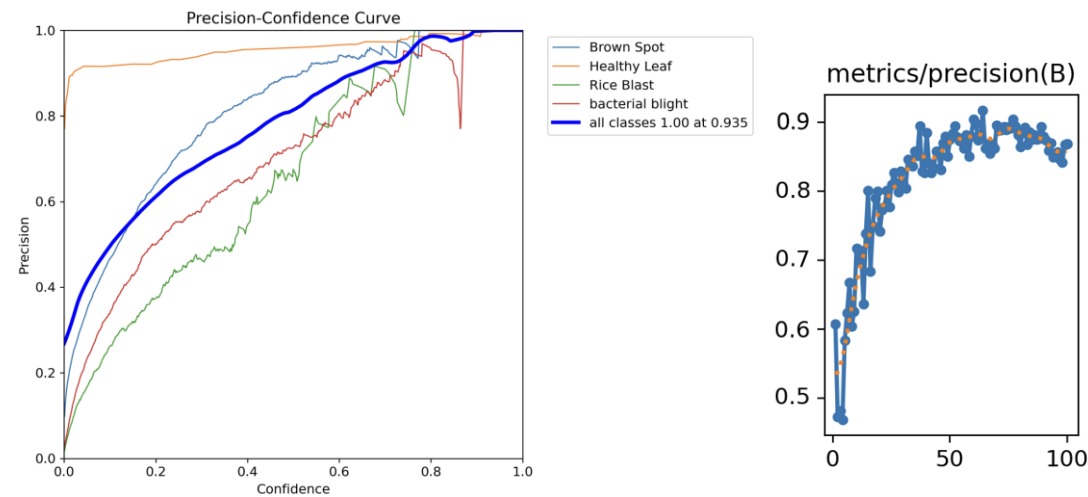


Figure 6.19 Precision-Confidence Curve based on the YOLOv11 model

6.4.2 Precision Recall

The YOLOv11 Precision-Recall Curve highlights the model’s ability to distinguish Brown Spot (0.918), Healthy Leaf (0.988), Rice Blast (0.901), and Bacterial Blight (0.907). Each colored line corresponds to a specific class, while the bold blue line combines all classes, achieving an mAP@0.5 of 0.929. Overall, Healthy Leaf exhibits the highest performance at 0.988, and the other diseases also maintain

solid precision-recall profiles. The tightly grouped lines near the top indicate that the model consistently balances precision (correct detections) and recall (total detections) across different confidence levels, indicating strong disease recognition capabilities.

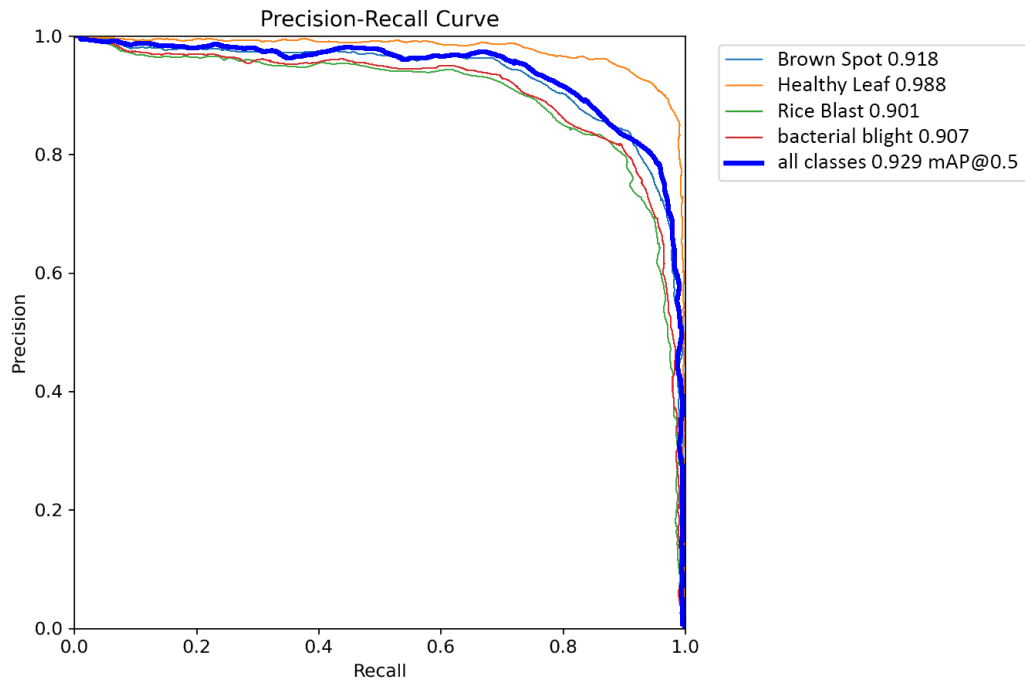


Figure 6.20 Precision-Recall Curve based on the YOLOv11 model

6.4.3 F1-Confidence Curve

The F1-Confidence Curve for the YOLOv11 model evaluates its performance in classifying rice leaf conditions, including Brown Spot, Healthy Leaf, Rice Blast, and Bacterial Blight. The x-axis represents the confidence threshold, and the y-axis indicates the F1 score, which measures the balance between precision and recall. The model achieves its optimal performance at a confidence threshold of 0.275, reaching an overall F1 score of 0.89, as indicated by the bold blue line. Among the individual classes, Healthy Leaf consistently exhibits the highest F1 scores across most confidence levels, showcasing the model's strong accuracy for this category. Brown Spot also shows robust performance, while Bacterial Blight and Rice Blast have comparatively lower F1 scores, indicating areas where the model's performance could

improve. This curve reflects YOLOv11's advancements in achieving higher overall accuracy compared to previous versions while identifying its strengths and weaknesses across different categories.

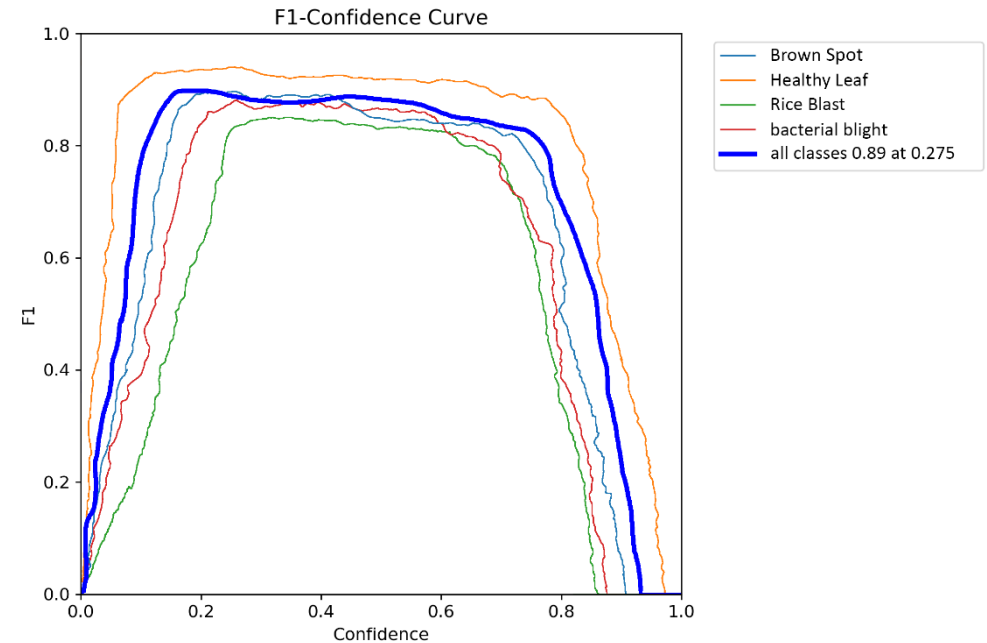


Figure 6.21 F1-Confidence Curve based on the YOLOv11 model

6.4.4 Confusion Metrics

The YOLOv11 confusion matrix shows how accurately the model differentiates between Brown Spot, Healthy Leaf, Rice Blast, Bacterial Blight, and background. The darker diagonal cells reflect correct classifications, while the off-diagonal cells reveal misclassifications. Brown Spot is correctly identified 89% of the time, with 11% mistakenly labeled as background. Healthy Leaf has the highest accuracy at 97%, leaving 3% incorrectly categorized as background. Rice Blast achieves 85% correctness, with 15% confused as background, and Bacterial Blight posts an 87% correct rate, with 13% mislabeled as background. Some background images are also being identified as diseases. Despite strong performance, these instances suggest there is still room for improvement, either through further model tuning or adding more training data.

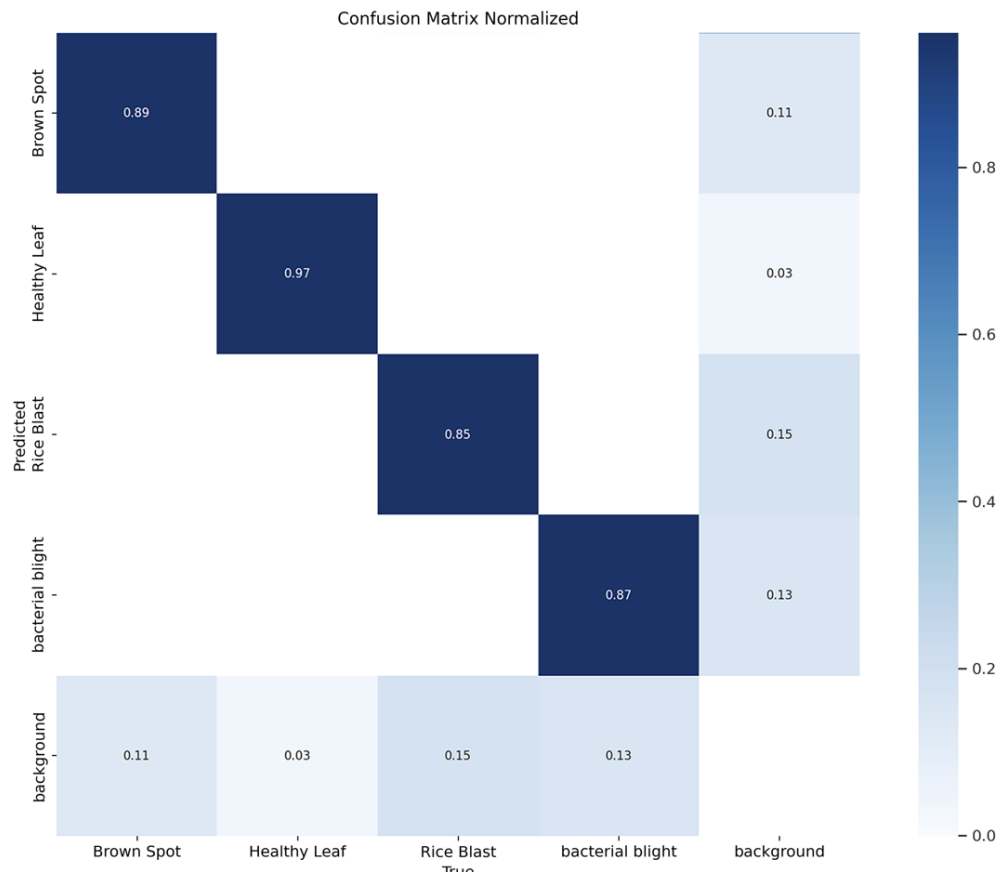


Figure 6.22 Confusion matrix based on the YOLOv11 model

6.4.5 Loss Metrics

The following YOLOv11 training graph shows how three loss types—box, classification, and distribution focal loss (DFL), decrease over 100 epochs. At the beginning, the losses are relatively high, indicating the model is still learning how to localize disease areas and classify them correctly. As training progresses, each loss steadily declines, reflecting improved performance in bounding-box placement and object recognition. By the end of the 100 epochs, the losses have stabilized at lower levels, suggesting that YOLOv11 has effectively learned to identify and detect rice leaf diseases.

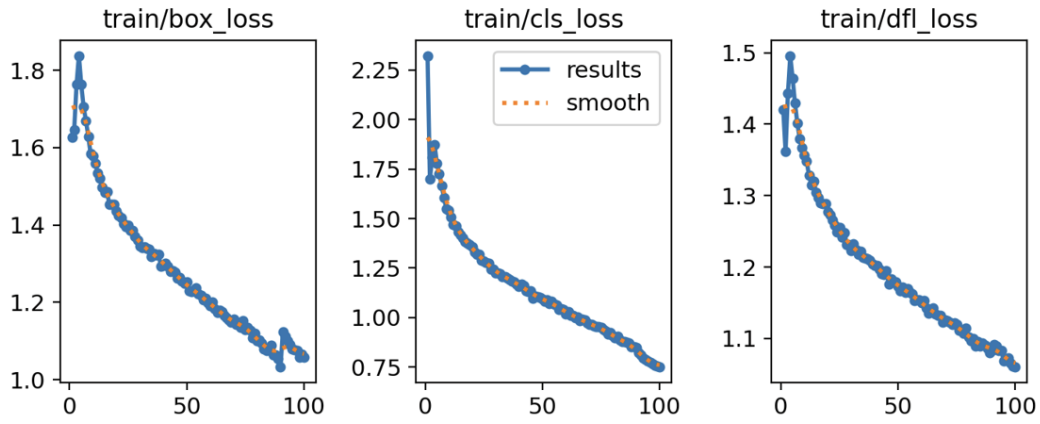


Figure 6.23 Loss metrics graph based on the YOLOv11 model

6.4.6 Results on Testing Images using the YOLOv11 Model

This section discusses the performance of the YOLOv11 model on testing images. The images given in the following were tested using the YOLOv11 model. Figures 6.21 to 6.2 display the detection results, including confidence levels, for all classes of rice leaf diseases.

Detection of Healthy Leaves

The detection results for healthy leaves are presented in Figure 6.21. The YOLOv11 model correctly identified the healthy leaves, highlighting its capability to differentiate between diseased and healthy plant parts.



Figure 6.24 Detection for Healthy Leaves

Detection of Bacterial Blight

Figure 6.22 illustrates the detection of bacterial leaf blight. The YOLOv11 model successfully localized and classified the bacterial blight diseases, showcasing its effectiveness in identifying this disease under various conditions.

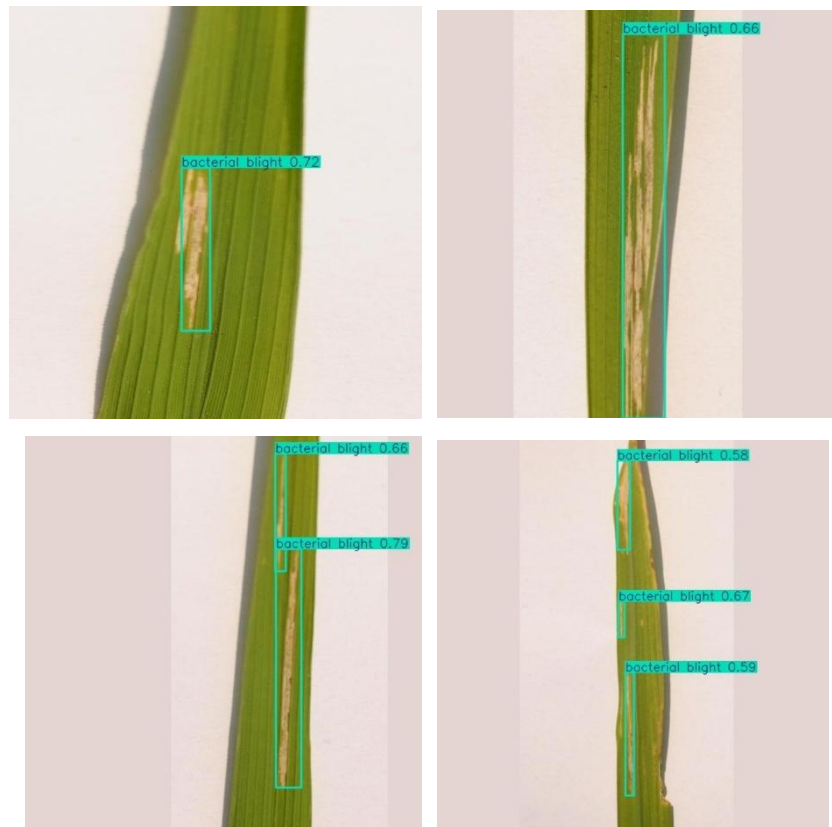


Figure 6.25 Detection for Bacterial Blights

Detection of Brown Spots

The YOLOv11 model successfully detected brown spot diseases on rice leaves, as shown in Figure 6.23. The model's predictions are marked with bounding boxes and corresponding confidence scores, demonstrating its ability to detect brown spot with high precision.

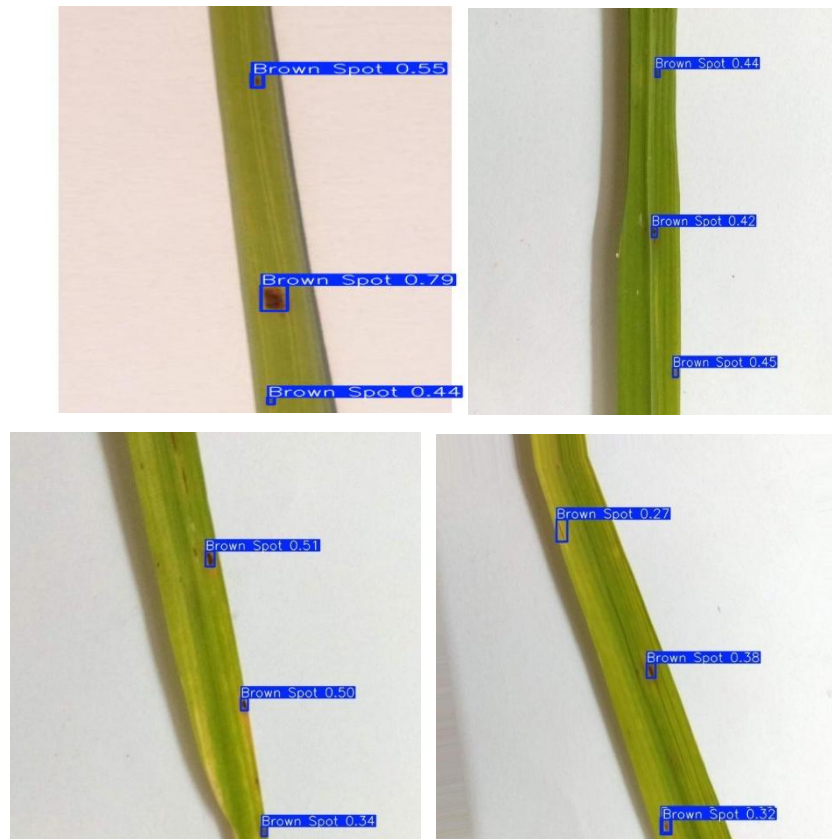


Figure 6.26 Detection for Brown Spots

Detection of Rice Blasts

The detection of blast disease on rice leaves is illustrated in Figure 6.24. The performance of the YOLOv11 model in detecting rice blasts is shown by the bounding boxes and confidence levels, indicating its robustness in identifying this specific disease.

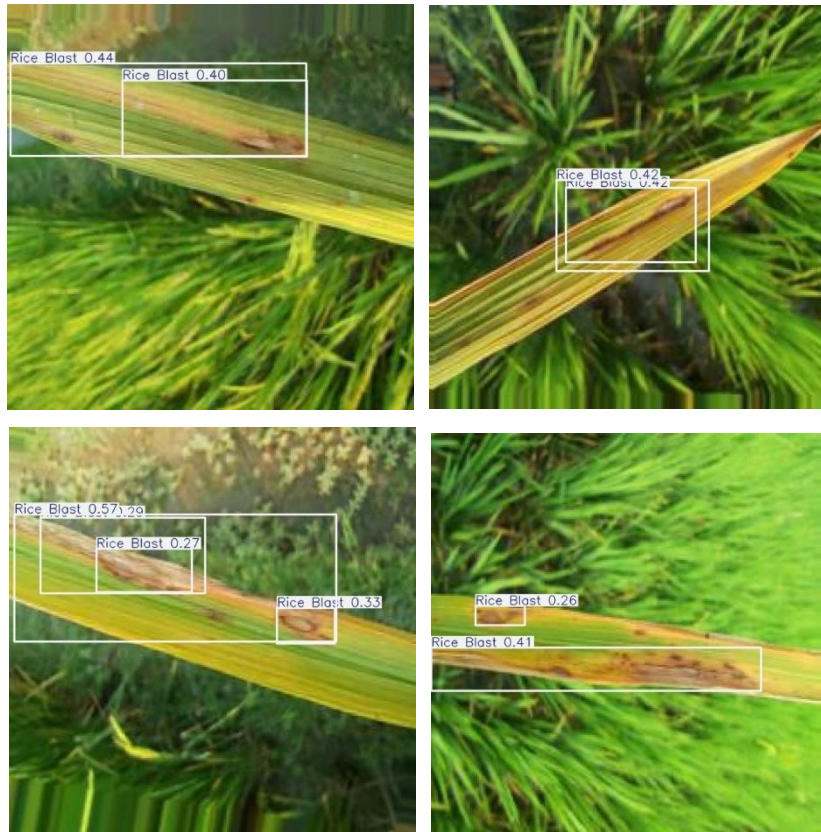


Figure 6.27 Detection for Rice Blasts

6.5 Comparisons of the Models:

This section provides a comparative analysis of the three models—YOLOv8, YOLOv10, and YOLOv11—developed in this research. The models were evaluated based on key performance metrics, including precision, F1-confidence scores, mean Average Precision (mAP), and inference time, to assess their strengths and weaknesses in detecting rice leaf diseases.

YOLOv8 achieved a precision of 88.0%, an F1-confidence score of 80%, and the fastest inference time of 10.2 milliseconds, making it ideal for real-time applications where speed is critical. YOLOv10 improved on this with a precision of 89.9%, an F1-confidence score of 85%, and an inference time of 12.5 milliseconds. YOLOv11 demonstrated the highest precision of 92.9% and an F1-confidence score of 89%, showcasing its ability to detect complex disease patterns with exceptional

accuracy. However, it required the longest inference time of 15.3 milliseconds, indicating a trade-off between accuracy and computational efficiency.

In summary, YOLOv11 is the most suitable for precision-focused applications where accuracy is paramount, such as detailed disease analysis. In contrast, YOLOv8 is better for scenarios requiring real-time detection due to its speed. YOLOv10 provides a balance between the two, offering improved accuracy and F1-confidence over YOLOv8 while maintaining relatively fast inference times.

6.6 Development and Analysis of Dashboard

6.6.1 Introduction to Dashboard

This section describes about the development of the dashboard that has been designed to provide a simple and interactive interface for the detection and analysis of rice leaf diseases based on the trained dataset. It incorporates YOLO models for precise and instant detection of diseases and has features for model comparison, dataset analytics, and disease information visualization.

6.6.2 Dashboard Summary

The Rice Leaf Disease Detection Dashboard is designed to provide a structured and intuitive experience for researchers, farmers, and agricultural enthusiasts. It seamlessly integrates data analytics, model evaluation, and real-time detection, allowing users to gain insights into rice leaf diseases and the effectiveness of YOLO-based machine learning models. The dashboard layout ensures a logical flow of information, starting from dataset analytics to model performance evaluation, comparison, and ultimately, an interactive image upload feature for real-time disease identification. By combining AI-driven insights with interactive functionalities, this tool aims to enhance disease detection accuracy and support precision agriculture.

The Dataset Analytics section provides a detailed overview of the dataset used to train and test the YOLO models. This section visually presents the number of images, annotations, and disease categories, ensuring that users understand how the AI has been trained. The Classes & Tags subsection further organizes the dataset, allowing researchers to assess the distribution of healthy and diseased leaves. This transparent approach builds confidence in the AI's reliability by allowing users to verify the comprehensiveness of the dataset, ensuring that it effectively represents real-world conditions.

To further evaluate the model's effectiveness, the Model Analytics section provides key performance metrics such as the confusion matrix and Precision-Confidence Curve. The confusion matrix reveals how accurately the models detect each disease while minimizing false positives and negatives, while the Precision-Confidence Curve helps users assess how well the models maintain accuracy across different confidence levels. These insights allow researchers to compare the strengths and weaknesses of YOLOv8, YOLOv10, and YOLOv11, ensuring they select the best model for their agricultural applications.

Finally, the interactive image upload and real-time detection feature makes this dashboard a practical tool for both researchers and farmers. Users can simply drag and drop rice leaf images in formats such as JPG and PNG to get an instant diagnosis of diseases. The dashboard also includes precision comparisons between different YOLO versions, providing a clear understanding of how each model performs in real-world scenarios. By combining AI-driven automation with an easy-to-use interface, this tool empowers agricultural specialists to make data-driven decisions, improve crop health, and enhance sustainability in rice farming.

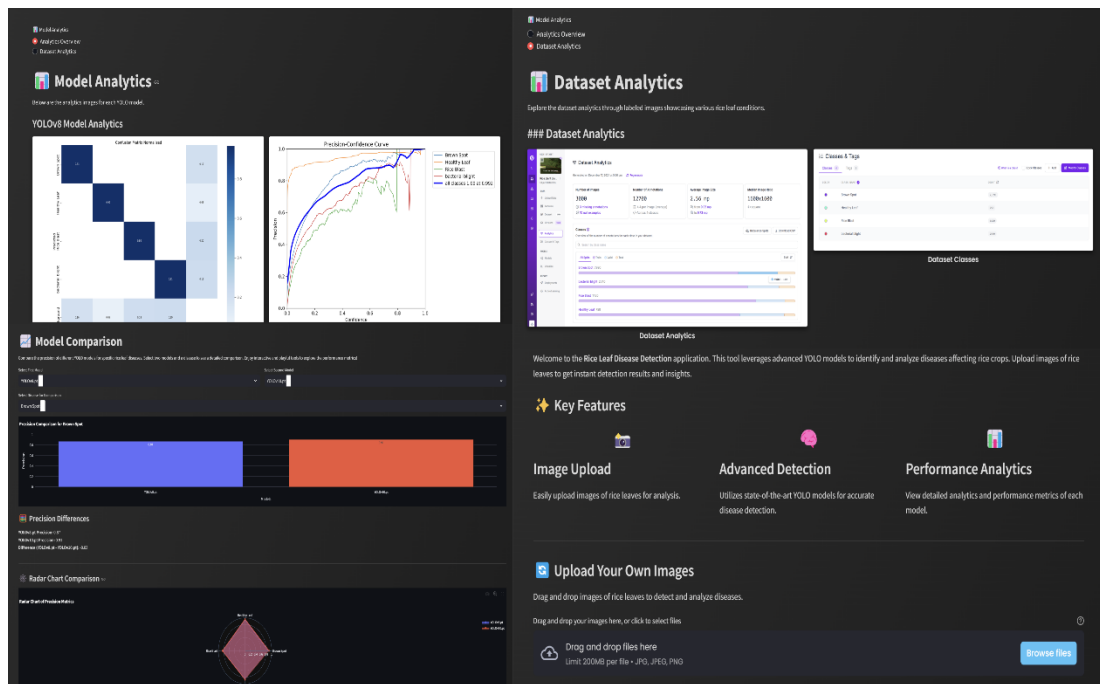


Figure 6.28 Rice Leaf Disease Detection Dashboard

6.7 Chapter Summary

This chapter explains the results of the YOLO models and describes the dashboard created for detecting rice leaf diseases. It highlights how YOLOv8, YOLOv10, and YOLOv11 performed. Misclassifications were analyzed using confusion matrices, and the models' training was shown to be effective through improved loss metrics. The dashboard, built with Streamlit, allows users to upload images, select models, and view results easily. This chapter shows how the models and dashboard can help users to know more about the detection models and rice leaf disease information.

CHAPTER 7

CONCLUSION AND RECOMMENDATIONS

7.1 Introduction

This chapter summarises the significant findings from the research on enhanced feature selection for identifying rice leaf diseases using different versions of YOLO algorithms. The study gives valuable insights into improving disease detection accuracy by analysing extensive datasets and comparing the performance of various versions of the YOLO models. This chapter provides a concise overview of the research findings, explores their importance, and suggests ways to improve the practicality and efficiency of the developed models in future studies.

7.2 Summary of the Key Findings:

This research effectively tackled the challenges of rice leaf disease detection by using advanced machine learning techniques, achieving all the research objectives. A high-quality dataset was developed as the foundation of this study, leveraging extensive exploratory data analysis and precise polygon annotation techniques. The dataset, comprising thousands of annotated images of rice leaves affected by bacterial blight, blast, and brown spot, ensured robust training for the detection models. This process not only enhanced the accuracy of disease-specific feature recognition but also addressed the inherent complexities of overlapping leaves and similar backgrounds in natural field images.

The study successfully developed and evaluated three custom models based on the YOLO framework—YOLOv8, YOLOv10, and YOLOv11. Among these, YOLOv11 emerged as the best-performing model, achieving the highest mean average precision (mAP) of 92.9% and an F1 score of 89%, reflecting its superior capability in

identifying and classifying rice leaf diseases. YOLOv10 and YOLOv8 also performed well, with mAP scores of 89.9% and 88.0% and F1 scores of 85% and 80%, respectively. These results highlight the advancements in detection accuracy and computational efficiency achieved by the enhanced YOLO-based models. Furthermore, the study revealed the effectiveness of using polygon labeling and data augmentation techniques in improving model generalization, enabling accurate predictions even under challenging conditions.

A significant contribution of this research is the development of the interactive web-based dashboard that integrates model analytics and real-time disease detection capabilities. The dashboard allows users to upload rice leaf images, visualize detection results, and access key insights about the models and their performance. This tool bridges the gap between advanced machine learning solutions and practical agricultural applications, offering actionable insights for farmers and agricultural stakeholders. By aligning its objectives with outcomes, the study demonstrates its potential to revolutionize rice disease management through early detection and precision diagnostics, ultimately contributing to sustainable agriculture and global food security.

7.3 Contributions to Knowledge

The research focused on developing and evaluating models with the YOLOv8, YOLOv10, and YOLOv11 algorithms for detecting and classifying rice leaf diseases. The YOLOv11 model demonstrated significant accuracy, achieving a mean average precision rate of 92.9%, highlighting its potential for enhancing disease detection processes in agriculture. The YOLOv08 model and YOLOv10 models also demonstrated significant accuracy, achieving a mean average precision rate of 88% and 89.9%. This project makes significant contributions to the field of agricultural technology by integrating advanced machine learning techniques with practical applications for detecting and managing rice leaf diseases. By developing a robust and interactive dashboard powered by YOLO models, it provides an efficient tool for real-time disease detection, offering precision and reliability across multiple disease types, including bacterial blight, brown spots, and rice blast. The study also enhances

understanding of YOLO model performance through detailed evaluations using metrics such as precision, recall, and confusion matrices, paving the way for further improvements in disease detection accuracy.

Furthermore, the dashboard simplifies complex detection processes, making it accessible to agricultural experts and farmers. By bridging the gap between advanced technology and practical agricultural needs, this project addresses critical challenges in crop management, ultimately contributing to improved yield quality and sustainability. The insights gained from this research offer a foundation for future studies and applications in precision agriculture, highlighting the transformative potential of AI-driven solutions in rice crop management.

7.4 Recommendations for Future Work

The exploration of agricultural disease detection using deep learning holds great promise for revolutionizing precision farming. By addressing the limitations identified in this study and building on its strengths, future advancements can significantly enhance the scalability, accuracy, and practicality of YOLO-based models. With a focus on real-world applications, these improvements could drive impactful changes in agricultural practices worldwide.

Expanding the dataset to include a wider variety of rice leaf conditions, environmental scenarios, and regional factors is a crucial step forward. Incorporating data that reflects different seasons, growth stages, and geographical contexts will improve the model's robustness and generalizability. Furthermore, integrating additional variables, such as weather patterns, soil quality, and pest activity, can provide a more comprehensive framework for disease detection, enabling predictions that account for complex agricultural ecosystems.

While this study centered on YOLOv8, YOLOv10, and YOLOv11 models, future research could benefit from exploring other algorithms, such as YOLOv7, YOLOv9, or hybrid models combining YOLO with emerging architectures like

Transformers or Convolutional Neural Networks (CNNs). These hybrid approaches could capture intricate spatial and contextual nuances, enhancing detection accuracy. Advanced techniques like 3D bounding boxes or multi-spectral imaging could also be explored to identify subtle disease traits more effectively. Additionally, integrating real-time data processing systems would significantly enhance the practical utility of the platform, ensuring predictions remain accurate and actionable in dynamic farming environments.

REFERENCES

- Aggarwal, S., Suchithra, M., Chandramouli, N., Sarada, M., Verma, A., Vetrithangam, D., Pant, B., & Adugna, B. A. (2022). Rice disease detection using artificial intelligence and machine learning techniques to improvise agro business. *Scientific Programming*, 2022, 1757888. <https://doi.org/10.1155/2022/1757888>
- Ahmed, K., Shahidi, T. R., Alam, S. M. I., & Momen, S. (2019). Rice Leaf Disease Detection Using Machine Learning Techniques. 2019 *International Conference on Sustainable Technologies for Industry 4.0 (STI)*, December 24–25. IEEE. <https://doi.org/10.1109/STI47673.2019.9068096>
- Agbulos, M. K., Sarmiento, Y., & Villaverde, J. (2021). Identification of Leaf Blast and Brown Spot Diseases on Rice Leaf with YOLO Algorithm. *Proceedings of the International Conference on Computer Science and Software Engineering*, 9545153. <https://doi.org/10.1109/ICCSSE52761.2021.9545153>
- Alif, M. A. R., & Hussain, M. (2024). YOLOv1 to YOLOv10: A comprehensive review of YOLO variants and their application in the agricultural domain. <https://doi.org/10.48550/arXiv.2406.10139>
- Asibi, A. E., Chai, Q., & Coulter, J. A. (2019). Rice Blast: A Disease with Implications for Global Food Security. *Agronomy*, 9(8), 451. <https://doi.org/10.3390/agronomy9080451>
- Bakirci, M. (2024). Utilizing YOLOv8 for enhanced traffic monitoring in intelligent transportation systems (ITS) applications. *Digital Signal Processing*, 152, September, Article 104594. <https://doi.org/10.1016/j.dsp.2024.104594>

- Bakirci, M., & Bayraktar, I. (2024). Boosting Aircraft Monitoring and Security through Ground Surveillance Optimization with YOLOv9. Presented at the *IEEE 12th International Symposium on Digital Forensics and Security (ISDFS)*, April 29. <https://doi.org/10.1109/ISDFS60797.2024.10527349>
- Canlas, C. J. N., Cortez, C. M. M., Dela Cruz, R. A. M., Padua, J. R. G., Timbol, A. G. G., Yumul, J. G., Trinidad, E. T., & Materum, L. (2021). Camera Radial Distance-Based Accuracy of a Bacterial Blight, Brown Spot, and Rice Blast Plant Disease Identification System for Remote Communications Deployment. *Journal of Image and Graphics*, 10(4), 158–165. <https://doi.org/10.18178/joig.10.4.158-165>
- Chen, C., Zheng, Z., Xu, T., Guo, S., Feng, S., Yao, W., & Lan, Y. (2023). YOLO Based UAV Technology: A Review of the Research and Its Applications. *Drones*, 7(3), 190. <https://doi.org/10.3390/drones7030190>
- Diwan, T., Ani, A., & Tembhurne, J. V. (2022). Object detection using YOLO: challenges, architectural successors, datasets and applications. *Multimedia Tools and Applications*, 82(6). <https://doi.org/10.1007/s11042-022-13644-y>
- Haque, M. E., Rahman, A., Junaid, I., Paul, M., & Hoque, S. (2022). Rice Leaf Disease Classification and Detection Using YOLOv5. *arXiv*. <https://doi.org/10.48550/arXiv.2209.01579>
- Haque, Trinh, D. C., Mac, A. T., Dang, K. G., Nguyen, H. T., Nguyen, H. T., & Bui, T. D. (2024). Alpha-EIOU-YOLOv8: An improved algorithm for rice leaf disease detection. *Agricultural Engineering*, 6(1), 302–317. <https://doi.org/10.3390/agriengineering6010018>
- International Rice Research Institute (IRRI). (n.d.). Rice production. Retrieved from <http://www.knowledgebank.irri.org/ericeproduction/welcome.htm>

- Ismail, U. I., Ahmed, M. K., Adam, M., & Salihu, S. (2023). Deep Learning Based Model for Rice Plant Disease Classification Using YOLOv8. *International Journal of Computer Science and Explorer (IJCSE)*, 1(1), 1–17. <https://doi.org/10.0710/IJCSE.2023840085>
- John, M., Bankole, I. A., Ajayi-Moses, O., Ijila, T., Jeje, T., & Patil, L. P. (2023). Relevance of Advanced Plant Disease Detection Techniques in Disease and Pest Management for Ensuring Food Security and Their Implication: A Review. *American Journal of Plant Sciences*, 14(11), November. <https://doi.org/10.4236/ajps.2023.1411086>
- Kaur, A., Guleria, K., & Trivedi, N. K. (2021). Rice Leaf Disease Detection: A Review. 2021 *International Conference on Signal Processing, Computing and Control (ISPCC)*, October 7–9. IEEE. <https://doi.org/10.1109/ISPCC53510.2021.9609473>
- Khanam, R., & Hussain, M. (2024). YOLOv11: An overview of the key architectural enhancements. <https://doi.org/10.48550/arXiv.2410.17725>
- Kiratiratanapruk, K., Temniranrat, P., Kitvimonrat, A., Sinthupinyo, W., & Patarapuwadol, S. (2020). Using Deep Learning Techniques to Detect Rice Diseases from Images of Rice Fields. *Conference Paper*. https://doi.org/10.1007/978-3-030-55789-8_20
- Liu, X., Wang, Y., Chen, X., & Chen, F. (2024). Real-time object detection method based on YOLOv5 and efficient mobile network. *Journal of Real-Time Image Processing*, 21, Article 56. <https://doi.org/10.1109/AIHCIR61661.2023.00014>
- Masood, M. H., Saim, H., Taj, M., & Awais, M. M. (2020). Early Disease Diagnosis for Rice Crop. Paper presented at the *ICLR 2020 Workshop on Computer Vision for Agriculture (CV4A)*. <https://doi.org/10.48550/arXiv.2004.04775>

- Morbekar, A., Parihar, A., & Jadhav, R. (2020). Crop Disease Detection Using YOLO. *2020 International Conference for Emerging Technology (INCET)*, June 5–7. IEEE. <https://doi.org/10.1109/INCET49848.2020.9153986>
- Nalley, L., Tsiboe, F., Durand-Morat, A., Shew, A., & Thoma, G. (2016). Economic and Environmental Impact of Rice Blast Pathogen (*Magnaporthe oryzae*) Alleviation in the United States. *PLOS ONE*. <https://doi.org/10.1371/journal.pone.0167295>
- Neupane, N., & Bhusal, K. (2021). A Review of Blast Disease of Rice in Nepal. *Journal of Plant Pathology & Microbiology*, 12(1), 1–5. <https://doi.org/10.35248/2157-7471.20.12.528>
- Niones, J. T., Sharp, R. T., Donayre, D. K. M., Oreiro, E. G. M., Milne, A. E., & Oliva, R. (2021). Dynamics of bacterial blight disease in resistant and susceptible rice varieties. *European Journal of Plant Pathology*, 160(4), 901–912. <https://doi.org/10.1007/s10658-021-02452-z>
- Rathore, Y. K., Janghel, R. R., Pandey, S. K., Kumar, A., Singh, K. U., & Shah, M. A. (2022). A novel AI-based approach for better segmentation of the fungal and bacterial leaf diseases of rice plant. *Journal of Advanced Research*, 1–12. <https://doi.org/10.1155/2022/6871085>
- Redmon, J., Divvala, S., Girshick, R., & Farhadi, A. (2015). You Only Look Once: Unified, Real-Time Object Detection. *arXiv*. <http://arxiv.org/abs/1506.02640>
- Roboflow. (2023). Improve Accuracy: Polygon Annotations for Object Detection. Retrieved from <https://blog.roboflow.com/polygon-vs-bounding-box-computer-vision-annotation/>
- Sanya, D. R. A., Syed-Ab-Rahman, S. F., Jia, A., Onésime, D., Kim, K.-M., Ahohuendo, B. C., & Rohr, J. R. (2022). A review of approaches to control bacterial leaf blight in rice. *World Journal of Microbiology and Biotechnology*, 38(7). <https://doi.org/10.1007/s11274-022-03298-1>

- Shuai, F., Qian, H., Wang, H., & Wang, W. (2024). Real-time object detection method based on YOLOv5 and efficient mobile network. *Journal of Real-Time Image Processing*, 21(2), March 2024. <https://doi.org/10.1007/s11554-024-01433-9>
- Sindhu, T. S., Chandrika, G., Divya, G., Sumathi, R., & Durga, P. (2023). Rice leaf disease detection using machine learning technique. *AIP Conference Proceedings*, 5(1), 0198893. <https://doi.org/10.1063/5.0198893>
- Surendhar, M., Anbuselvam, Y., & Ivin, J. J. S. (2021). Status of Rice Brown Spot (*Helminthosporium oryza*) Management in India: A Review. *Agricultural Reviews*. <https://doi.org/10.18805/ag.R-2111>
- Tian, L., Xue, B., Wang, Z., Li, D., Yao, X., Cao, Q., Zhu, Y., Cao, W., & Cheng, T. (2021). Spectroscopic detection of rice leaf blast infection from asymptomatic to mild stages with integrated machine learning and feature selection. *Remote Sensing of Environment*, 258, 112350. <https://doi.org/10.1016/j.rse.2021.112350>
- Upadhyay, S. K., & Kumar, A. (2021). Early-Stage Brown Spot Disease Recognition in Paddy Using Image Processing and Deep Learning Techniques. *Transactions on Emerging Telecommunications Technologies*, 38(6), 1713–1720. <https://doi.org/10.18280/ts.380619>
- Udayananda, V., Shyalika, C., & Pathirage, N. (2022). Rice plant disease diagnosing using machine learning techniques: a comprehensive review. *SN Applied Sciences*, 4(11), 5194. <https://doi.org/10.1007/s42452-022-05194-7>
- Vijayakumar, A., & Vairavasundaram, S. (2024). YOLO-based Object Detection Models: A Review and its Applications. *Multimedia Tools and Applications*. <https://doi.org/10.1007/s11042-024-18872-y>

- Viswanatha, V., Chandana, R. K., & Ramachandra, A. C. (2022). Real Time Object Detection System with YOLO and CNN Models: A Review. *Journal of Xi'an University of Architecture & Technology (JXAT)*, XIV(VII), July 2022. <https://doi.org/10.37896/JXAT14.07/315415>
- Wang, A., Chen, H., Liu, L., Chen, K., Lin, Z., Han, J., & Ding, G. (2024). YOLOv10: Real-time end-to-end object detection. <https://doi.org/10.48550/arXiv.2405.14458>
- Wang, J., Wang, H., & Wu, A. (2024). YOLOv5 for Enhanced Small Object Detection in Paired IR and Depth Images. In *Proceedings of the IEEE 7th Advanced Information Technology, Electronic and Automation Control Conference (IAEAC)*, March 15–17. IEEE. <https://doi.org/10.1109/IAEAC59436.2024.10503604>
- Wen, G., Li, M., Tan, Y., Shi, C., Luo, Y., & Luo, W. (2025). Enhanced YOLOv8 algorithm for leaf disease detection with lightweight GOCR-ELAN module and loss function: WSIoU. *Computers in Biology and Medicine*, 186, 109630. <https://doi.org/10.1016/j.compbiomed.2024.109630>
- Yang, G., Wang, J., Nie, Z., Yang, H., & Yu, S. (2023). A Lightweight YOLOv8 Tomato Detection Algorithm Combining Feature Enhancement and Attention. *Agronomy*, 13(7). <https://doi.org/10.3390/agronomy13071824>
- Zhang, C., Chen, X., Liu, P., He, B., Li, W., & Song, T. (2024). Automated detection and segmentation of tunnel defects and objects using YOLOv8-CM. *Tunneling and Underground Space Technology*, 134, 105857. <https://doi.org/10.1016/j.tust.2024.105857>
- Zhu, R., Hao, F., & Ma, D. (2023). Research on Polygon Pest-Infected Leaf Region Detection Based on YOLOv8. *Agriculture*, 13(12), 2253. <https://doi.org/10.3390/agriculture13122253>

